

PROJECT REPORT ON

SMART EV BATTERY MONITORING AND PROTECTION

Submitted in partial fulfilment of the requirements of the Summer Online Internship

on

EMBEDDED SYSTEMS



Submitted by

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Organized by

Andhra Pradesh State Skill Development Corporation (APSSDC)

Duration :8 weeks

2026-2027

CERTIFICATE

This is to certify that the project entitled "**SMART EV BATTERY MONITORING AND PROTECTION**" is a Bonafide work carried out by the following students of the Department of Electrical and Electronics Engineering, [Sri Venkateshwara College of Engineering, Tirupati](#) during the Summer online **Internship on Embedded Systems** organized by the **Andhra Pradesh State Skill Development Corporation (APSSDC)**.

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The project was completed under the guidance of **Mr. B. Mahidhar, Executive Technical, APSDDC**. The work presented in this report is original and has been carried out by the above students as part of the internship requirements.

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Internship Guide

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DECLARATION

We hereby declare that the project entitled "**SMART EV BATTERY MOINITORING AND PROTECTION**" submitted as part of the **Summer Online Internship on Embedded Systems** organized by the **Andhra Pradesh State Skill Development Corporation (APSSDC)** is our original work.

The project has been carried out under the guidance of **Mr. B. Mahidhar, Executive Technical, APSSDC**. We further declare that this report has not been submitted previously, either in whole or in part, for the award of any degree, diploma, or certificate by any other institution.

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Finally, we express our heartfelt gratitude to our parents, friends, and everyone who directly or indirectly supported us throughout the internship and motivated us to complete this project successfully.

Place: _____

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ABSTRACT

Smart EV Battery Monitoring and Protection is a system designed to improve the safety, reliability, and performance of electric vehicle (EV) battery packs by continuously monitoring important battery parameters. The system measures battery voltage, current, temperature, and State of Charge (SOC) using sensors and a microcontroller. The collected data is displayed on an LCD/OLED display and can also be monitored in real time through IoT platforms for remote access.

To ensure battery safety, the system provides protection against overcharging, over-discharging, overheating, and overcurrent conditions. When any parameter exceeds the safe operating limit, the controller activates a relay to disconnect the battery from the load or charging source and alerts the user through a buzzer or warning message. This helps prevent battery damage, extends battery life, and improves overall system efficiency.

The proposed system is cost-effective, easy to implement, and suitable for educational and prototype EV applications. It demonstrates how intelligent battery monitoring and protection can enhance the performance, safety, and lifespan of electric vehicle battery systems

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CHAPTER 1

INTRODUCTION TO EMBEDDED SYSTEMS

CHAPTER 1

INTRODUCTION TO EMBEDDED SYSTEMS

1.1 Introduction

An embedded system is a specialized computing system designed to perform a dedicated function or a set of specific functions within a larger electronic or mechanical system. Unlike general-purpose computers, which are capable of executing a wide range of applications, embedded systems are developed to perform predetermined tasks with high efficiency, reliability, and accuracy. They consist of both hardware and software components that work together to achieve real-time control and monitoring of devices. The hardware generally includes a microcontroller or microprocessor, memory units, input/output interfaces, communication modules, sensors, and actuators, while the software consists of embedded firmware programmed to execute the desired operations.

Structure of an Embedded System

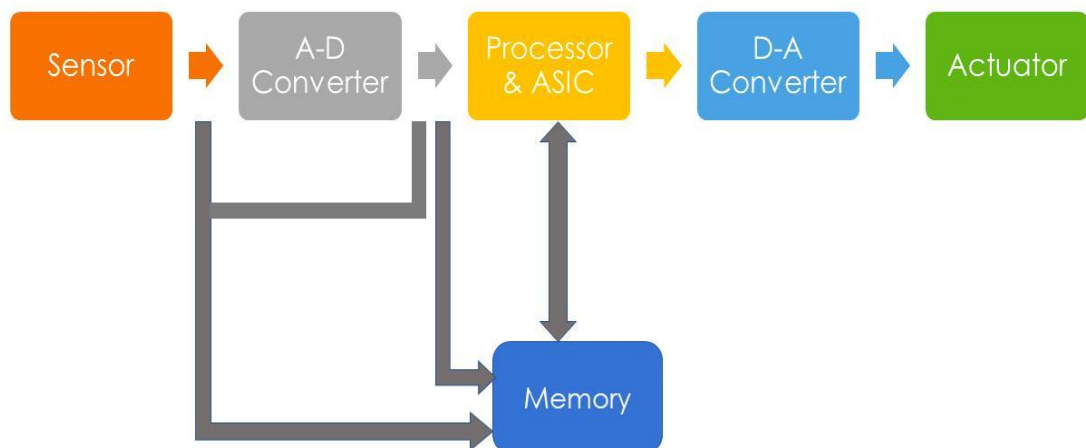


Figure 1.1 Basic Block Diagram of an Embedded System

The evolution of embedded systems has transformed modern technology by making electronic devices smarter, faster, and more efficient. Initially, embedded systems were developed for simple control applications, but advancements in semiconductor technology

and software development have enabled them to perform complex operations involving artificial intelligence, wireless communication, cloud connectivity, and real-time data processing. Today, embedded systems are an integral part of almost every industry, including automotive, healthcare, industrial automation, consumer electronics, aerospace, agriculture, robotics, renewable energy, and smart home applications.

Embedded systems are designed with specific objectives such as low power consumption, compact size, cost effectiveness, high performance, and long-term reliability. Most embedded systems operate continuously without human intervention and are expected to respond immediately to external events. This real-time capability makes them suitable for applications where timing and accuracy are critical. For example, in automotive systems, embedded controllers continuously monitor engine parameters, braking systems, battery management systems, and safety mechanisms to ensure efficient vehicle operation and passenger safety.

In recent years, the rapid growth of Electric Vehicles (EVs) has significantly increased the importance of embedded systems in battery monitoring and energy management. Modern EVs rely on embedded controllers to monitor battery voltage, current, temperature, and State of Charge (SOC) while protecting the battery from abnormal operating conditions such as overcharging, deep discharge, excessive current, and overheating. These intelligent systems enhance battery safety, improve energy efficiency, extend battery lifespan, and provide real-time information to users and maintenance personnel.

The project titled "**Smart EV Battery Monitoring and Protection**" is an application of embedded systems that demonstrates how a microcontroller-based system can continuously monitor battery health and automatically protect the battery from unsafe conditions. By integrating sensors, display units, protection circuits, and intelligent control algorithms, the proposed system provides a reliable and economical solution for efficient battery management. The implementation of such embedded technologies contributes to the development of safer, smarter, and more sustainable electric transportation systems.

1.2 Characteristics of Embedded Systems

Embedded systems possess several unique characteristics that distinguish them from conventional computing systems. These characteristics make them suitable for performing

dedicated tasks with high precision, efficiency, and reliability. Since embedded systems are developed for specific applications, both hardware and software are optimized to achieve maximum performance while minimizing cost, power consumption, and physical size.

One of the most important characteristics of an embedded system is real-time operation. Many embedded applications require immediate responses to external events within predefined time limits. A delay in processing can lead to system failure or safety hazards, especially in applications such as medical equipment, industrial automation, aerospace systems, and electric vehicle battery management systems.

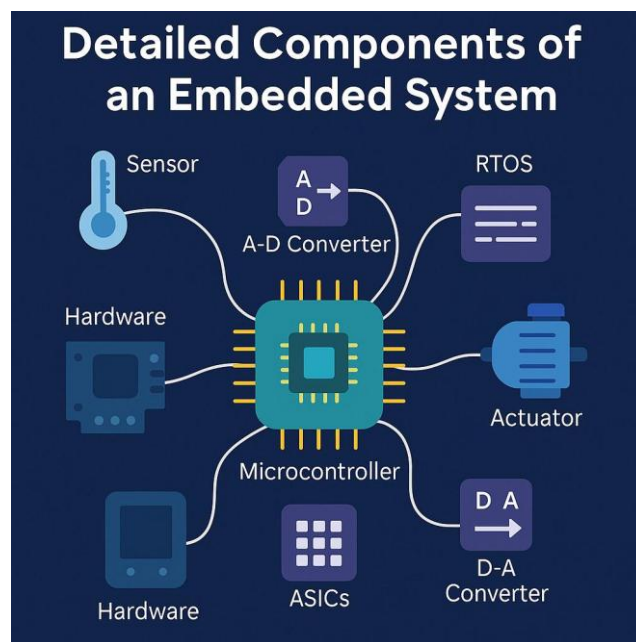


Figure 1.2 Components of an Embedded System

Another significant characteristic is reliability. Embedded systems are expected to operate continuously for long periods without failure. Since they are often used in mission-critical applications, they must provide consistent performance under varying environmental conditions such as temperature changes, electrical disturbances, and mechanical vibrations.

Low power consumption is another essential feature of embedded systems. Many embedded devices are battery-operated and therefore require efficient power management techniques to maximize battery life. Designers implement power-saving modes, optimized software algorithms, and energy-efficient hardware components to reduce overall power consumption.

Embedded systems are also designed to be compact and lightweight, allowing them to be integrated into portable devices and space-constrained applications. The use of System-on-Chip (SoC) technology and highly integrated microcontrollers has enabled manufacturers to develop powerful embedded systems within very small dimensions.

Another important characteristic is application-specific functionality. Unlike personal computers that can execute multiple software applications, embedded systems are programmed to perform one or a limited number of dedicated tasks. This specialization improves execution speed, reduces memory requirements, and enhances overall system efficiency.

Finally, embedded systems emphasize high performance with low cost. Designers carefully select hardware components and optimize software to achieve the required functionality while minimizing production costs. This balance between performance and affordability has made embedded systems widely adopted across various industries.

1.3 Components of an Embedded System

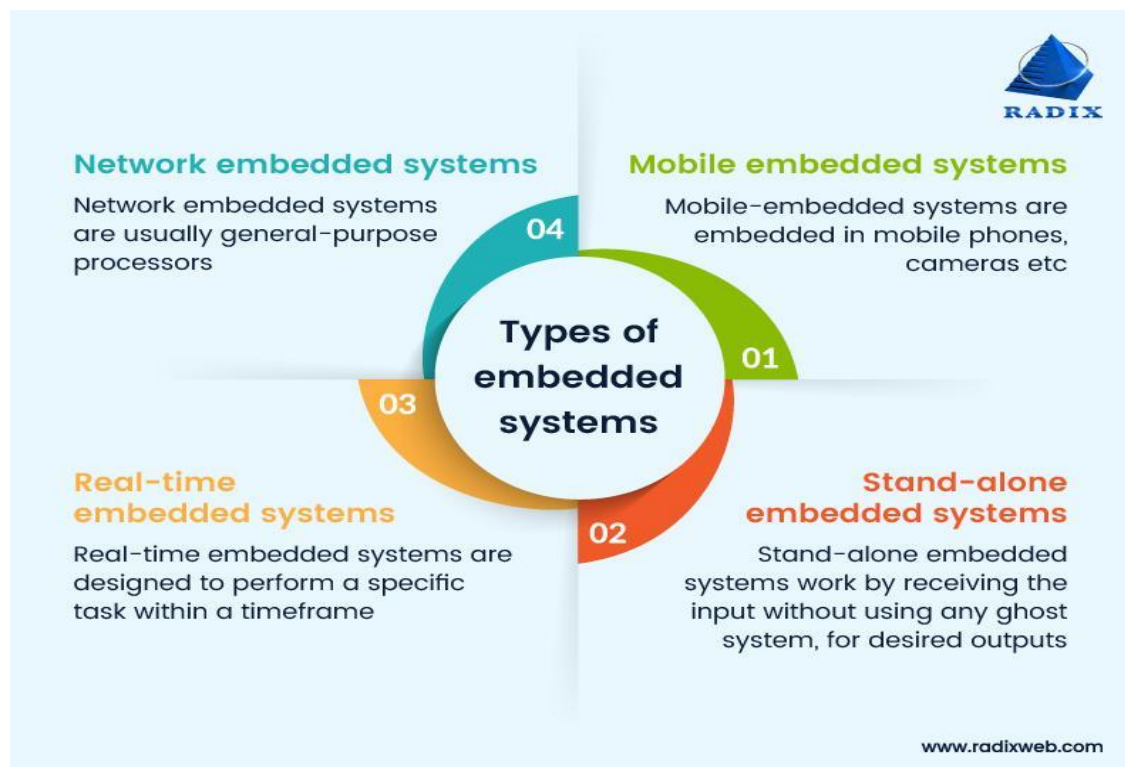


Figure 1.3 Classification of Embedded Systems

An embedded system consists of several hardware and software components that work together to perform a specific task. The efficiency and reliability of an embedded system largely depend on the proper integration of these components.

The **microcontroller or microprocessor** serves as the brain of the embedded system. It executes the program instructions stored in memory and controls all connected peripherals. Modern microcontrollers integrate CPU, memory, timers, communication interfaces, and input/output ports on a single chip, making them ideal for embedded applications.

The **memory unit** stores both program instructions and runtime data. Read Only Memory (ROM), Flash Memory, or EEPROM stores the firmware permanently, while Random Access Memory (RAM) temporarily stores variables and intermediate calculations during execution.

Sensors are input devices that detect physical parameters such as temperature, voltage, current, pressure, light intensity, humidity, and motion. They convert physical quantities into electrical signals that can be processed by the microcontroller.

Actuators are output devices that convert electrical signals into physical actions. Examples include relays, motors, solenoid valves, buzzers, LEDs, and cooling fans. In this project, the relay disconnects the battery under fault conditions while the buzzer alerts the user about abnormal operating conditions.

Communication modules such as UART, SPI, I²C, CAN, Bluetooth, Wi-Fi, and GSM enable embedded systems to exchange data with external devices or cloud platforms. These interfaces support remote monitoring and control, which are becoming increasingly important in modern smart applications.

The **power supply unit** provides stable operating voltage to all system components. Voltage regulators and protection circuits ensure reliable operation despite fluctuations in the input supply.

embedded systems to exchange data with external devices or cloud platforms. These interfaces support remote monitoring and control, which are becoming increasingly important in modern smart applications.

Finally, the **embedded software or firmware** coordinates the operation of all hardware components. The firmware is usually developed using Embedded C or Arduino programming and includes algorithms for sensing, data processing, decision-making, communication, and fault protection..

1.4 Types of Embedded Systems

Embedded systems are classified into different categories based on their functionality, performance, and application requirements.

Stand-alone Embedded Systems operate independently without requiring a host computer. They receive input from sensors or users, process the information internally, and generate the required output. Examples include microwave ovens, washing machines, calculators, digital cameras, and smart home appliances.

Real-Time Embedded Systems are designed to respond within strict timing constraints. These systems are further classified as hard real-time and soft real-time systems. Hard real-time systems cannot tolerate delays because failure to respond within the specified time can lead to severe consequences. Airbag control systems and aircraft navigation systems are examples of hard real-time systems. Soft real-time systems allow small delays without causing system failure.

Networked Embedded Systems communicate with other embedded devices through wired or wireless communication networks. They are widely used in industrial automation, smart grids, Internet of Things (IoT), and intelligent transportation systems.

Mobile Embedded Systems are portable devices powered by rechargeable batteries. Smartphones, wearable fitness trackers, GPS navigation devices, drones, and handheld medical equipment are examples of mobile embedded systems. These systems emphasize energy efficiency, compact design, and wireless connectivity.

Modern embedded systems often combine multiple categories to deliver advanced functionality. For example, an Electric Vehicle Battery Management System is a real-time, networked, and mobile embedded system that continuously monitors battery parameters while communicating with other vehicle control units

Product Applications

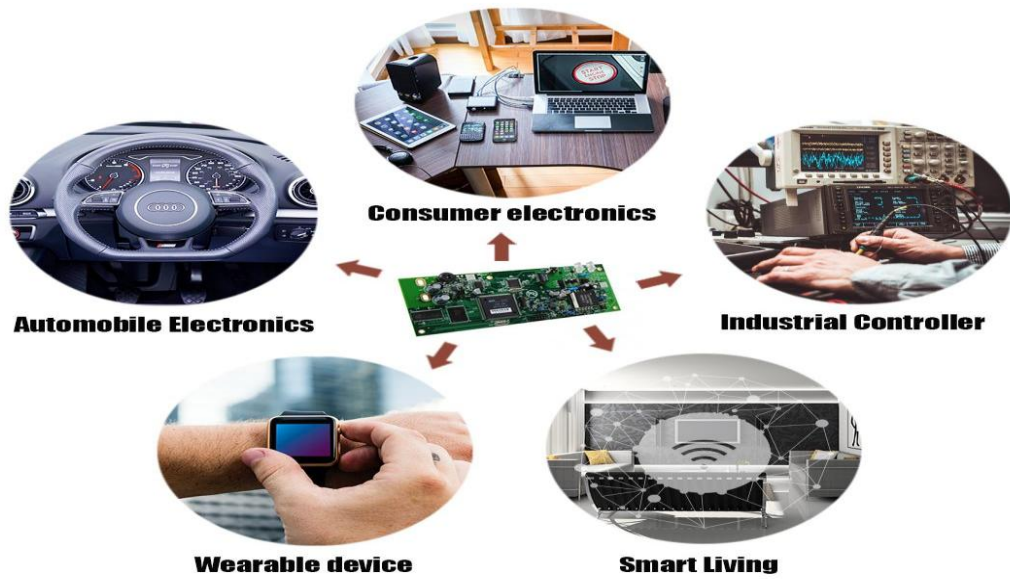


Figure 1.4 Applications of Embedded Systems in Various Industries

1.5 Applications of Embedded Systems

Embedded systems have become an essential part of modern technology due to their ability to provide intelligent control, automation, and real-time monitoring across various industries.

In the automotive industry, embedded systems control engine management, anti-lock braking systems (ABS), airbag deployment, battery management systems (BMS), electric power steering, infotainment systems, and advanced driver assistance systems (ADAS). These applications improve vehicle safety, fuel efficiency, and overall driving experience.

In healthcare, embedded systems are used in medical devices such as ECG monitors, patient monitoring systems, infusion pumps, insulin delivery systems, digital thermometers, blood pressure monitors, and portable diagnostic equipment.

The consumer electronics sector relies heavily on embedded systems in televisions, washing machines, refrigerators, microwave ovens, digital cameras, smartphones, printers, and gaming consoles.

In industrial automation, embedded systems control programmable logic controllers (PLCs), robotic arms, conveyor systems, CNC machines, process monitoring equipment, and energy management systems.

The renewable energy sector uses embedded systems for solar charge controllers, wind turbine monitoring, smart inverters, battery energy storage systems, and electric vehicle charging stations.

The Internet of Things (IoT) integrates embedded systems with cloud computing and wireless communication, enabling smart homes, smart agriculture, environmental monitoring, wearable devices, and smart cities.

In this project, embedded systems play a crucial role in continuously monitoring battery parameters, displaying real-time information, generating warning alerts, and automatically protecting the EV battery from unsafe operating conditions.

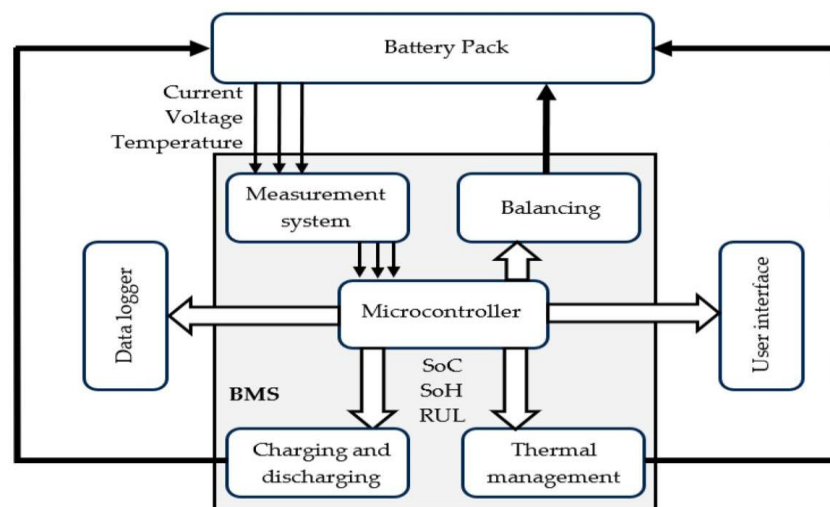


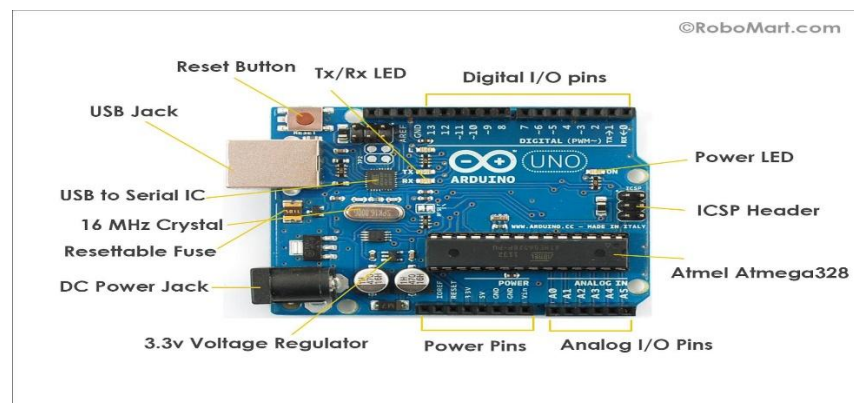
Figure 1.5 Embedded System Used in an Electric Vehicle Battery Management System

1.6 Advantages of Embedded Systems

Embedded systems offer numerous advantages, making them indispensable in modern electronic products. They provide high processing efficiency while consuming minimal power, making them suitable for battery-operated devices. Their compact size allows easy integration into small electronic equipment without increasing system complexity.

These systems are highly reliable because they are designed for dedicated applications and undergo extensive testing before deployment. Their low production cost makes them economically feasible for mass production, while their fast response time enables real-time decision-making in safety-critical applications. Additionally, embedded systems require minimal maintenance and can operate continuously for years without significant performance degradation.

Figure 1.6 Arduino UNO Development Board



1.7 Challenges of Embedded Systems

Despite their advantages, embedded systems also face several challenges. Limited memory and processing resources restrict the implementation of computationally intensive algorithms. Developing reliable embedded software requires extensive testing to ensure system stability under all operating conditions.

Power management remains a major challenge in portable embedded devices. Designers must balance performance and energy efficiency to maximize battery life. Hardware failures, software bugs, and cybersecurity threats can also affect system reliability. As embedded systems become increasingly connected through IoT technologies, protecting them from unauthorized access and cyberattacks has become an important design consideration.

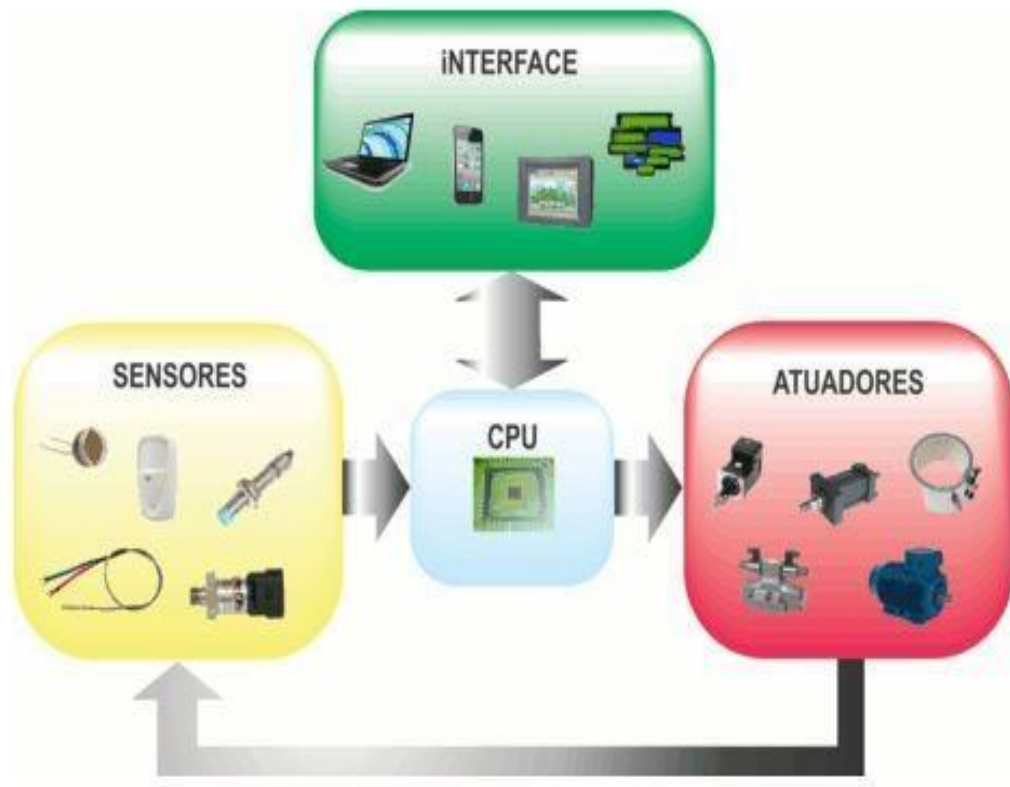


Figure 1.7 General Architecture of an Embedded System

1.8 Summary

Embedded systems have become the backbone of modern electronic technology by providing intelligent, reliable, and efficient control for dedicated applications. Their ability to perform real-time monitoring, data processing, and automated decision-making has revolutionized industries such as automotive, healthcare, industrial automation, and renewable energy. This chapter introduced the concept, characteristics, components, classifications, applications, advantages, and challenges of embedded systems. Understanding these fundamentals provides the necessary background for developing advanced embedded applications such as the Smart EV Battery Monitoring and Protection system. The next chapter discusses the microcontroller architecture and its role in implementing the proposed battery monitoring and protection system.

CHAPTER 2

INTRODUCTION TO MICROCONTROLLER

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INTRODUCTION TO MICROCONTROLLER

2.1 Introduction

A microcontroller is a compact and integrated electronic device that contains a processor, memory, and input/output peripherals on a integrated circuit (IC). It is specifically designed to perform dedicated control tasks in embedded systems. Unlike a microprocessor, which requires external memory and peripheral devices to function, a microcontroller integrates all the essential components into one chip, making it an ideal solution for compact, cost-effective, and power-efficient applications. Due to these advantages, microcontrollers have become the backbone of modern embedded systems and are extensively used in automation, consumer electronics, industrial control, medical equipment, robotics, home appliances, renewable energy systems, and electric vehicle

Block diagram of uC

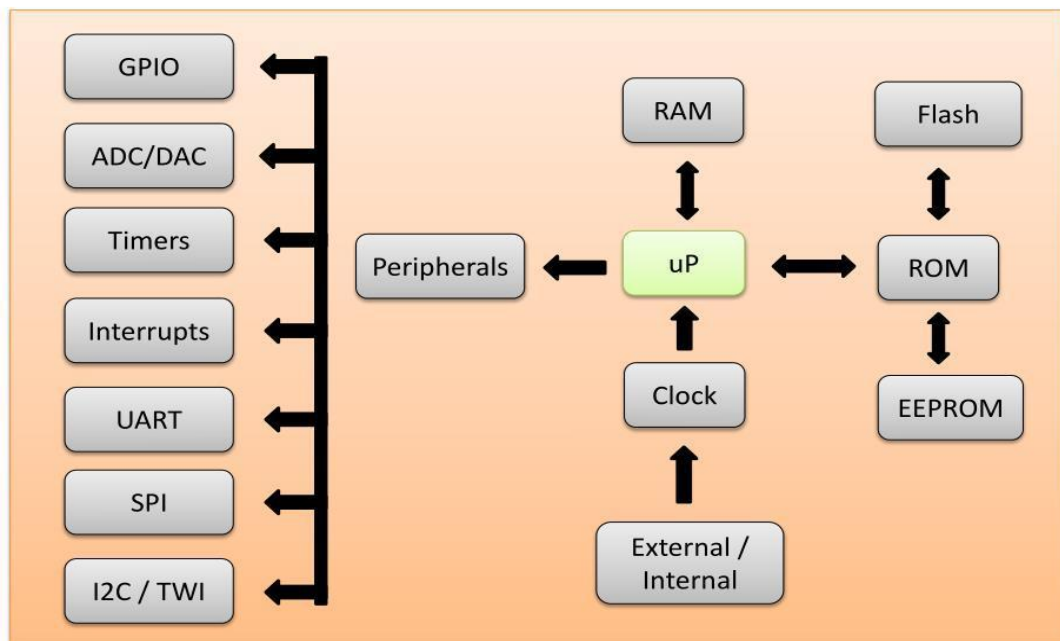


Figure 2.1 Internal Architecture of a Microcontroller

In recent years, the rapid advancement of electric vehicle technology has highlighted the importance of microcontrollers in Battery Management Systems (BMS). A microcontroller continuously measures battery voltage, current, temperature, and State of Charge (SOC), analyzes these parameters, and initiates protection mechanisms whenever abnormal conditions occur. By performing these tasks accurately and efficiently, microcontrollers improve battery safety, enhance system reliability, and extend battery life. In the proposed **Smart EV Battery Monitoring and Protection** project, the Arduino Uno microcontroller serves as the central processing unit, coordinating sensor readings, protection logic, relay control, LCD display updates, and alarm generation.

2.2 History of Microcontrollers

The concept of microcontrollers emerged during the early 1970s with the development of integrated circuit technology. Before microcontrollers were introduced, electronic control systems relied on multiple integrated circuits, making them expensive, bulky, and difficult to maintain. The introduction of microcontrollers revolutionized embedded system design by integrating the processor, memory, timers, and input/output ports into a single chip.

The first commercially available microcontroller, the **Intel 8048**, was introduced in 1976 and was primarily used in keyboards, industrial controllers, and communication devices.

It was followed by the **Intel 8051**, which became one of the most successful and widely used microcontrollers in embedded applications. Over the years, companies such as

Microchip Technology, Atmel, Texas Instruments, STMicroelectronics, and NXP Semiconductors introduced advanced microcontrollers with improved processing

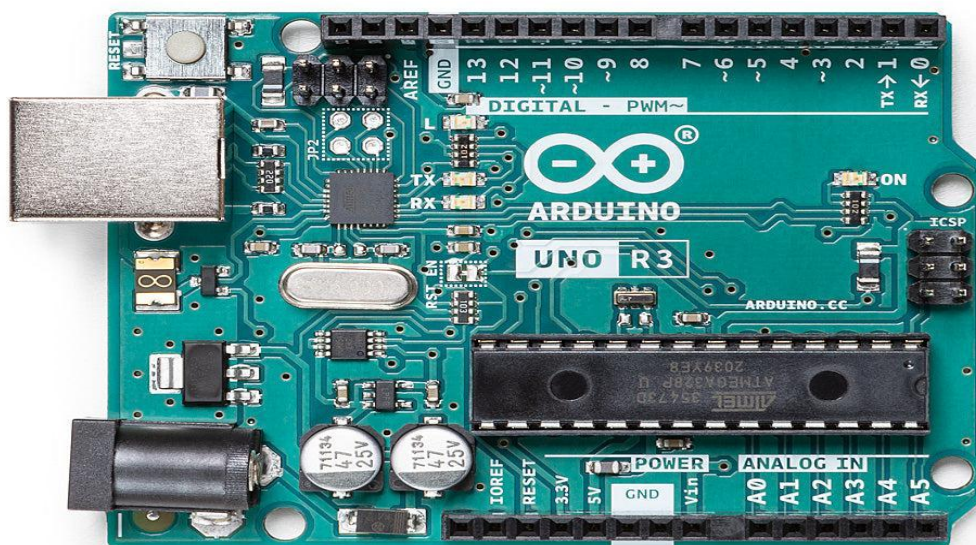


Figure 2.2 Arduino Uno Development Board (ATmega328P)

Today, modern microcontrollers feature high-speed processors, multiple analog-to-digital converters (ADCs), communication protocols such as UART, SPI, I²C, CAN, USB, Wi-Fi, and Bluetooth, making them suitable for Internet of Things (IoT), industrial automation, and smart electric vehicle applications.

2.3 Architecture of a Microcontroller

The architecture of a microcontroller represents the internal organization of its functional components and how they interact with one another. Although different manufacturers implement different architectures, most modern microcontrollers contain similar fundamental blocks.

The Central Processing Unit (CPU) is responsible for executing instructions and controlling the operation of the entire system. The memory subsystem consists of Flash memory for storing the program code, RAM for temporary data storage, and EEPROM for storing permanent configuration parameters. Input and Output (I/O) ports provide communication between the microcontroller and external devices such as sensors, displays, relays, and switches.

The Analog-to-Digital Converter (ADC) converts analog signals from sensors into digital values that can be processed by the CPU. Timers and counters perform time measurement, event counting, pulse generation, and PWM signal generation. Communication modules enable serial communication using protocols such as UART, SPI, and I²C. Interrupt controllers allow the processor to respond immediately to external events without continuously polling input devices.

Together, these components form an efficient embedded control system capable of performing real-time monitoring and decision-making.

2.4 Components of a Microcontroller

A microcontroller consists of several integrated hardware modules that work together to perform embedded control functions.

2.4.1 Central Processing Unit (CPU)

The CPU is the brain of the microcontroller. It executes instructions stored in program memory, performs arithmetic and logical operations, controls peripheral devices, and manages communication between hardware components.

2.4.2 Program Memory (Flash ROM)

Flash memory stores the embedded software permanently. Unlike RAM, the contents of Flash memory remain even when power is removed, allowing the microcontroller to execute the same program whenever it is powered on.

2.4.3 Random Access Memory (RAM)

RAM is used for temporary storage of variables, sensor data, intermediate calculations, and stack operations during program execution.

2.4.4 EEPROM

EEPROM is non-volatile memory used for storing calibration values, configuration parameters, and user settings that should remain unchanged even after power is switched off.

2.4.5 Input and Output Ports

Digital I/O ports enable the microcontroller to communicate with external devices. Inputs receive signals from switches and sensors, while outputs control LEDs, relays, buzzers, motors, and displays.

2.4.6 Analog-to-Digital Converter (ADC)

Most physical sensors produce analog voltages. The ADC converts these voltages into digital numbers that can be processed by the CPU. In this project, voltage, current, and temperature sensors utilize the ADC channels of the Arduino Uno.

2.4.7 Timers and Counters

Timers generate precise time delays and PWM signals, while counters count external pulses. They are essential for scheduling periodic tasks and controlling hardware timing.

2.4.8 Communication Interfaces

Modern microcontrollers support multiple communication protocols including:

- UART
- SPI
- I²C
- CAN Bus
- USB

These interfaces enable communication with sensors, displays, wireless modules, and other embedded controllers.

2.5 Arduino Uno Microcontroller

The Arduino Uno is one of the most popular development boards used in embedded systems education, research, and prototype development. It is based on the **ATmega328P**

microcontroller and provides a simple hardware platform with an easy-to-use programming environment. Its low cost, open-source ecosystem, and extensive community support make it ideal for beginners as well as professional developers.

The Arduino Uno operates at 5 V and includes 14 digital I/O pins, six analog input pins, six PWM outputs, timers, serial communication interfaces, and a USB interface for programming. It is capable of reading analog sensor values, controlling digital outputs, and communicating with external modules.

In the **Smart EV Battery Monitoring and Protection** project, the Arduino Uno performs the following functions:

- Reads battery voltage using voltage sensors.
- Measures charging and discharging current.
- Monitors battery temperature.
- Calculates State of Charge (SOC).
- Displays battery parameters on the LCD.
- Controls the protection relay.
- Activates the buzzer during fault conditions.

2.6 Arduino IDE and Embedded Programming

The Arduino Integrated Development Environment (IDE) is a software platform used to write, compile, and upload programs to Arduino boards. It provides a simple interface that supports C/C++ programming with built-in libraries for sensor interfacing, display control, communication protocols, and peripheral management.

The Arduino programming model consists mainly of two functions:

Setup() initializes the hardware and executes only once after power-up.

Loop() runs continuously, repeatedly executing the programmed logic throughout system operation.

For this project, the program continuously reads sensor values, processes battery parameters, checks protection limits, updates the LCD, controls the relay, and activates warning alarms whenever abnormal conditions are detected.

2.7 Applications of Microcontrollers

Microcontrollers are used in a wide variety of applications due to their flexibility, reliability, and low cost.

Some important applications include:

- Electric Vehicle Battery Management Systems
- Home Automation Systems
- Smart Energy Meters
- Industrial Process Control
- Medical Monitoring Equipment
- Consumer Electronics
- Agricultural Automation
- Robotics
- Renewable Energy Systems
- Security and Surveillance Systems
- Traffic Signal Controllers
- IoT-Based Smart Devices

The increasing adoption of Industry 4.0 and smart manufacturing technologies continues to expand the role of microcontrollers in modern engineering applications.

2.8 Advantages of Microcontrollers

Microcontrollers provide several significant advantages in embedded system design. Their compact architecture reduces hardware complexity and overall system cost. They consume very little power, making them suitable for battery-operated applications. Integrated peripherals eliminate the need for multiple external components, improving reliability while reducing PCB size.

Microcontrollers also offer high-speed processing, real-time operation, simple programming, excellent flexibility, and long operational life. Their ability to interface with numerous sensors and communication modules makes them suitable for intelligent automation systems, including electric vehicle battery monitoring and protection.

2.9 Summary

Microcontrollers are the fundamental building blocks of embedded systems and play a vital role in modern electronic applications. Their integrated architecture, low power consumption, high reliability, and real-time processing capabilities make them ideal for automation and control systems. This chapter discussed the history, architecture, internal components, Arduino Uno, programming environment, applications, and advantages of microcontrollers. These concepts provide the technical foundation required to implement the proposed **Smart EV Battery Monitoring and Protection** system, where the Arduino

CHAPTER 3

LITERATURE SURVEY

CHAPTER 3

LITERATURE SURVEY

3.1 Introduction

The rapid growth of Electric Vehicles (EVs) has significantly increased the demand for efficient Battery Management Systems (BMS) that ensure battery safety, reliability, and long operational life. The battery is the most critical and expensive component of an electric vehicle, contributing nearly 30–40% of the total vehicle cost. Since lithium-ion batteries are highly sensitive to operating conditions, continuous monitoring and protection are essential for maintaining optimal performance and preventing failures. Battery monitoring systems continuously measure important parameters such as voltage, current, temperature, and State of Charge (SOC) to ensure safe operation under varying load conditions.

A literature survey provides an overview of existing research, technologies, and methodologies related to battery monitoring and protection systems. It helps identify the strengths and limitations of current approaches while revealing research gaps that can be addressed through improved designs. Researchers across the world have proposed various Battery Management Systems incorporating embedded controllers, IoT technology, wireless communication, cloud computing, machine learning, and artificial intelligence to improve battery monitoring accuracy and reliability.

This chapter reviews the existing technologies and research work related to Electric Vehicle Battery Management Systems, battery monitoring techniques, protection methods, and intelligent monitoring solutions. The review also highlights the limitations of current systems and explains how the proposed project provides a simple, economical, and effective solution suitable for educational and prototype EV applications.

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computing, machine learning, and artificial intelligence to improve battery monitoring accuracy

3.2 Electric Vehicle Battery Management System (BMS)

A Battery Management System (BMS) is an electronic control system that manages rechargeable battery packs by monitoring battery parameters, protecting the battery from unsafe operating conditions, estimating battery health, and optimizing charging and discharging performance. The BMS acts as the "brain" of the battery pack and is responsible for maintaining the battery within its safe operating limits.

Modern Battery Management Systems continuously monitor cell voltage, battery current, temperature, State of Charge (SOC), and State of Health (SOH). These parameters help determine the battery's remaining energy, charging efficiency, and overall health condition. If abnormal conditions such as overcharging, over-discharging, overheating, or excessive current are detected, the BMS immediately disconnects the battery or limits current flow to prevent damage.

In electric vehicles, the BMS also communicates with the vehicle control unit, charging system, motor controller, and dashboard display. Advanced Battery Management Systems incorporate cell balancing, fault diagnosis, thermal management, wireless communication, and cloud-based monitoring to improve battery safety and extend operational life.

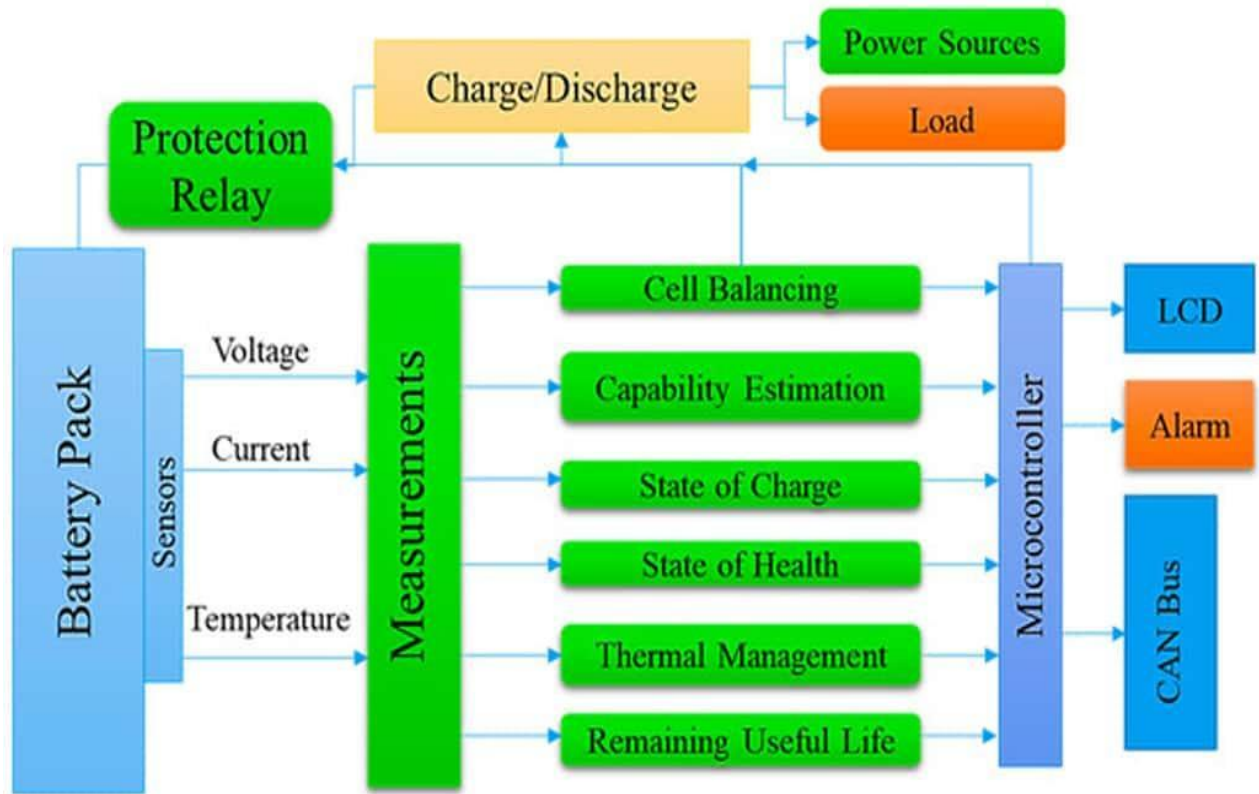


Figure 3.1 Basic Architecture of a Battery Management System (BMS)

3.3 Battery Monitoring Techniques

Battery monitoring is one of the most important functions of a Battery Management System. Researchers have developed numerous techniques to accurately monitor battery performance under different operating conditions.

The simplest monitoring method involves direct measurement of battery voltage using voltage divider circuits and analog-to-digital converters. Voltage measurement provides an estimate of battery condition and remaining charge.

Current monitoring is commonly performed using Hall Effect sensors or shunt resistors. Continuous measurement of charging and discharging current allows accurate estimation of battery capacity, power consumption, and energy efficiency.

Temperature monitoring is essential because excessive battery temperature can significantly reduce battery life and may even lead to thermal runaway. Temperature sensors such as LM35, DS18B20, thermistors, and digital temperature sensors are widely used for continuous battery temperature measurement.

Modern Battery Management Systems estimate State of Charge (SOC) using techniques such as:

- Coulomb Counting
- Open Circuit Voltage Method
- Kalman Filter Algorithm
- Artificial Neural Networks
- Machine Learning Models

Among these methods, Coulomb Counting and Open Circuit Voltage techniques are commonly used due to their simplicity and reasonable accuracy.

3.4 Battery Protection Techniques

Battery protection is essential for ensuring safe battery operation and preventing irreversible damage. Researchers have proposed various protection mechanisms to safeguard lithium-ion batteries against abnormal operating conditions.

Overvoltage protection prevents battery charging beyond the maximum allowable voltage. Excessive charging can cause overheating, electrolyte decomposition, and permanent battery damage.

Undervoltage protection prevents deep discharge by disconnecting the battery when the voltage falls below a predefined threshold. Deep discharge reduces battery capacity and shortens battery lifespan.

Overcurrent protection limits excessive charging or discharging current that may damage battery cells or electronic components. Protection circuits typically use current sensors combined with relay control or electronic switching devices such as MOSFETs.

Temperature protection monitors battery temperature continuously and disconnects the battery whenever excessive heat is detected. Cooling fans, heat sinks, and thermal management systems are also integrated into advanced battery packs to maintain safe operating temperatures.

Many commercial Battery Management Systems implement multiple protection mechanisms simultaneously to ensure maximum safety under all operating conditions.

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3.5 IoT-Based Battery Monitoring Systems

The Internet of Things (IoT) has transformed traditional battery monitoring systems into intelligent remote monitoring platforms. IoT-based Battery Management Systems collect real-time battery data and transmit it to cloud servers through wireless communication technologies such as Wi-Fi, GSM, Bluetooth, ZigBee, or LoRa.

Users can monitor battery voltage, current, temperature, charging status, and State of Charge using smartphones, tablets, or web applications from any location. Cloud-based storage also enables historical data analysis, predictive maintenance, and battery performance optimization.

Several researchers have integrated IoT platforms such as Blynk, ThingSpeak, Firebase, MQTT, and AWS IoT to develop smart battery monitoring solutions. These systems improve maintenance efficiency by allowing technicians to detect battery faults before they become critical failures.

Although IoT-based monitoring significantly enhances system intelligence, it increases

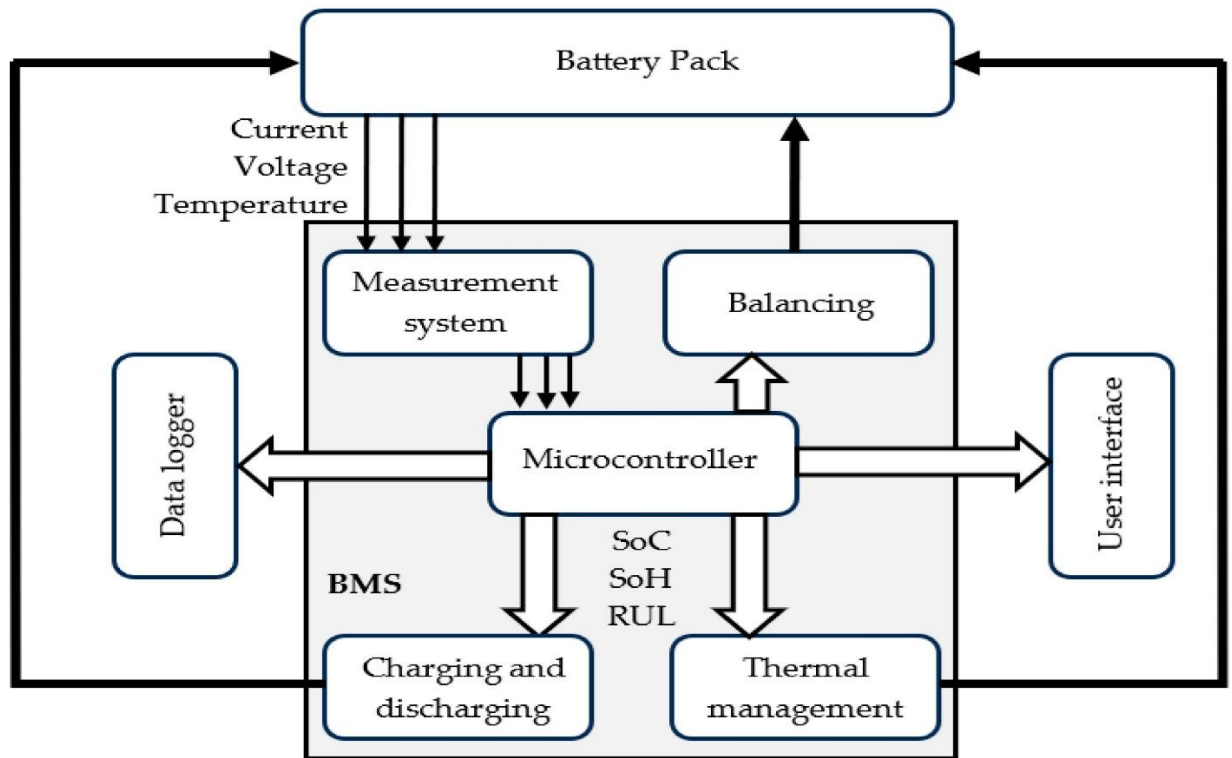


Figure 3.2 Electric Vehicle Battery Pack Integrated with Battery Management System

3.6 Review of Existing Research

Paper 1: IoT-Based Battery Monitoring System

Researchers developed an IoT-enabled Battery Management System capable of monitoring battery voltage, current, and temperature remotely through cloud-based applications. The system provides real-time alerts whenever battery parameters exceed safe operating limits. The study demonstrated improved monitoring convenience but required continuous internet connectivity for reliable operation.

Paper 2: Intelligent Battery Management Using Arduino

This research presented an Arduino-based Battery Management System for small-scale electric vehicles. The system measured battery voltage, current, and temperature while

displaying real-time information on an LCD display. Relay-based protection was incorporated to disconnect the battery under fault conditions. The study concluded that low-cost embedded systems are suitable for educational and prototype applications.

Paper 3: Lithium-Ion Battery Protection System

Researchers proposed an intelligent battery protection circuit using current sensors, temperature sensors, and microcontroller-based decision-making. The protection algorithm successfully prevented overcharging, deep discharge, excessive current, and overheating. Experimental results showed significant improvement in battery lifespan and operational safety.

Paper 4: Battery State of Charge Estimation

Several researchers developed State of Charge estimation algorithms using Kalman Filters, Coulomb Counting, Artificial Neural Networks, and Machine Learning techniques. Advanced estimation methods provided higher accuracy but required increased computational resources and more complex implementation compared to conventional techniques.

Paper 5: Smart Electric Vehicle Battery Monitoring

This research integrated multiple sensors with wireless communication to continuously monitor battery performance in electric vehicles. Data were displayed locally and transmitted to cloud servers for remote supervision. The proposed system demonstrated improved battery reliability and reduced maintenance requirements through continuous monitoring and automatic fault detection.

3.7 Research Gap

Although numerous Battery Management Systems have been developed, several limitations still exist in existing solutions.

Many commercial systems are expensive and unsuitable for educational institutions or prototype development. Advanced Battery Management Systems often require

sophisticated hardware, complex software algorithms, and cloud infrastructure, increasing implementation cost.

Some low-cost monitoring systems provide parameter measurement but lack automatic protection mechanisms. Other systems provide protection but do not support continuous monitoring or real-time user feedback. Many IoT-based solutions depend on stable internet connectivity, which may not always be available.

Furthermore, several systems focus only on battery monitoring without integrating comprehensive safety features such as relay-based isolation, audible alarms, and real-time local display.

These limitations highlight the need for an economical, reliable, and easy-to-implement battery monitoring system that combines continuous monitoring with automatic protection.

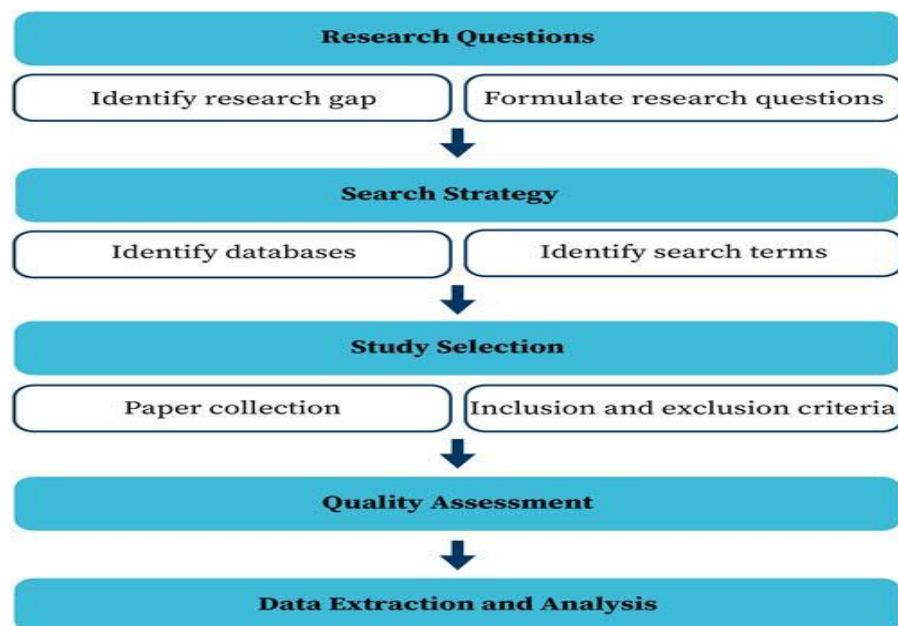


Figure: 3.3 Literature Survey Flow

3.8 Proposed System

To overcome the limitations identified in previous research, this project proposes a **Smart EV Battery Monitoring and Protection System** based on the Arduino Uno microcontroller.

The proposed system continuously monitors battery voltage, charging/discharging current, battery temperature, and State of Charge (SOC). The measured parameters are displayed on an LCD screen, allowing users to monitor battery performance in real time.

Whenever any battery parameter exceeds the predefined safety limit, the Arduino immediately activates a relay to disconnect the battery and generates an audible warning through a buzzer. This automatic protection prevents battery damage caused by overvoltage, undervoltage, overheating, and excessive current.

Compared to existing systems, the proposed design offers the following advantages:

- Low implementation cost
- Simple hardware design
- Easy programming using Arduino IDE
- Real-time battery monitoring
- Automatic protection using relay control
- LCD-based parameter display
- Expandable IoT capability
- Suitable for educational, research, and prototype applications

3.9 Summary

This chapter reviewed the existing literature related to Electric Vehicle Battery Management Systems, battery monitoring techniques, protection methods, and IoT-based monitoring technologies. Various research papers demonstrated the importance of continuously monitoring battery voltage, current, temperature, and State of Charge while implementing protection mechanisms to ensure battery safety. The survey also identified several limitations in current Battery Management Systems, including high cost, hardware complexity, and limited protection capabilities. Based on these observations, the proposed **Smart EV Battery Monitoring and Protection** system was developed as a simple, reliable, cost-effective, and scalable solution that integrates continuous monitoring with

CHAPTER 4

INTRODUCTION TO THE PROJECT

CHAPTER 4

INTRODUCTION TO THE PROJECT

4.1 Introduction

The increasing adoption of Electric Vehicles (EVs) has created a strong demand for intelligent systems capable of ensuring battery safety, improving energy efficiency, and extending battery life. The battery is the heart of an electric vehicle, and its performance directly influences the driving range, charging time, reliability, and overall efficiency of the vehicle. Since EV batteries operate under varying environmental and load conditions, continuous monitoring of battery parameters is essential to prevent failures and maintain safe operation. A Battery Management System (BMS) plays a crucial role by monitoring battery voltage, current, temperature, and State of Charge (SOC), while also providing protection against abnormal conditions such as overcharging, deep discharge, overheating, and excessive current.

The project titled "Smart EV Battery Monitoring and Protection" is designed to develop a microcontroller-based battery monitoring system that continuously measures critical battery parameters and automatically protects the battery whenever unsafe conditions are detected. The system utilizes an Arduino Uno as the central controller, interfaced with voltage, current, and temperature sensors to acquire real-time battery information. The measured parameters are processed by the microcontroller, displayed on an LCD, and used to determine the operational status of the battery.

To enhance battery safety, the system incorporates automatic protection mechanisms using a relay module and a buzzer. Whenever battery voltage, current, or temperature exceeds predefined safety limits, the relay disconnects the battery from the load or charging source while the buzzer alerts the user. This intelligent monitoring and protection mechanism minimizes battery damage, improves reliability, and extends battery lifespan. The proposed system is cost-effective, easy to implement, and suitable for educational institutions, research laboratories, prototype EV applications, and small-scale energy storage systems.

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effective, easy to implement, and suitable for educational institutions, research laboratories, prototype EV applications, and small-scale energy.

4.2 Problem Statement

Electric vehicle batteries are highly sensitive to operating conditions such as voltage fluctuations, excessive current flow, and temperature variations. Improper battery management may result in reduced battery capacity, overheating, permanent cell damage, shortened battery life, or even fire hazards in severe cases. Conventional battery monitoring methods often provide limited information and require manual inspection, making it difficult to detect faults at an early stage.

Many commercial Battery Management Systems are expensive and incorporate complex hardware and software architectures, making them unsuitable for educational purposes and prototype development. Therefore, there is a need for an economical, reliable, and intelligent battery monitoring system capable of continuously monitoring battery parameters while automatically protecting the battery from unsafe operating conditions.

4.3 Need for the Project

The rapid expansion of electric mobility has increased the demand for efficient battery monitoring technologies that ensure safe battery operation and improve overall vehicle performance. Battery failures caused by overcharging, over-discharging, overheating, and excessive current significantly affect vehicle reliability and maintenance costs. Continuous battery monitoring enables early detection of abnormal operating conditions, allowing preventive action before permanent damage occurs.

The proposed project addresses these challenges by providing a simple yet effective Battery Management System that integrates real-time monitoring, automatic protection, and user-friendly parameter display. The project also serves as an educational platform for understanding embedded systems, sensor interfacing, battery protection techniques, and microcontroller programming.

4.4 Objectives

The primary objectives of the proposed project are:

- To continuously monitor battery voltage, current, temperature, and State of Charge (SOC).
- To display battery parameters in real time using an LCD display.
- To automatically detect abnormal battery operating conditions.
- To protect the battery against overvoltage, undervoltage, overcurrent, and overheating.
- To disconnect the battery using a relay during unsafe conditions.
- To generate warning alerts through a buzzer.
- To improve battery safety and operational reliability.
- To develop a low-cost and user-friendly Battery Management System suitable for educational and prototype applications.

4.5 Scope of the Project

The scope of this project includes the design and implementation of a microcontroller-based Battery Monitoring and Protection System for electric vehicle battery packs. The project focuses on measuring essential battery parameters, processing sensor data, displaying real-time information, and implementing automatic protection mechanisms.

Although the present work is developed as a prototype using Arduino Uno, the proposed architecture can be extended for commercial electric vehicles by incorporating wireless communication, cloud monitoring, mobile applications, battery health estimation, and advanced battery balancing techniques. The project also provides an excellent platform for learning embedded system design, sensor interfacing, and battery management technologies.

4.6 Proposed System

The proposed system consists of an Arduino Uno microcontroller connected to voltage, current, and temperature sensors. These sensors continuously measure battery parameters and transmit analog signals to the microcontroller through its Analog-to-Digital Converter (ADC).

The Arduino processes the acquired data, calculates battery State of Charge (SOC), compares measured values with predefined safety thresholds, and updates the LCD display. Under normal operating conditions, the relay remains energized, allowing normal charging and discharging operations. Whenever abnormal conditions are detected, the relay disconnects the battery while the buzzer generates an audible warning.

This integrated monitoring and protection mechanism improves battery safety, enhances operational efficiency, and minimizes maintenance requirements.

4.7 Block Diagram

The proposed system consists of the following functional blocks:

- Battery
- Voltage Sensor
- Current Sensor
- Temperature Sensor
- Arduino Uno
- LCD Display
- Relay Module
- Buzzer
- Power Supply

The sensors continuously send battery data to the Arduino. The Arduino processes this information and controls the LCD, relay, and buzzer according to battery operating conditions.

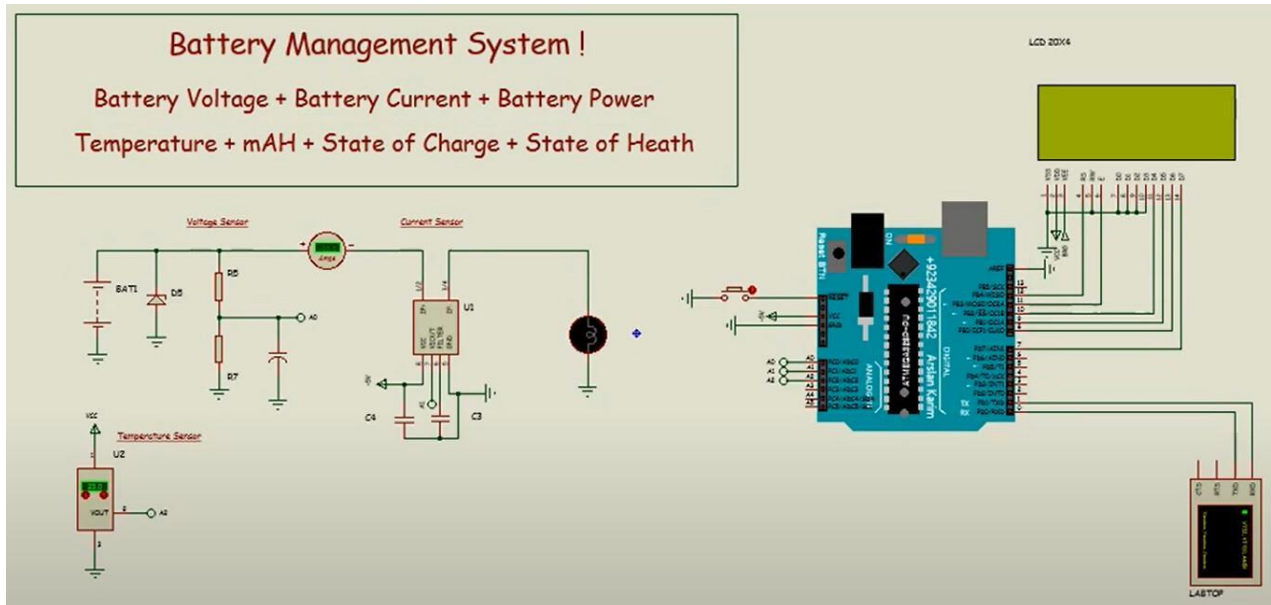


Figure 4.1: Block Diagram of the Proposed Smart EV Battery Monitoring and Protection System

4.8 Working Principle

Initially, the Arduino initializes all sensors, LCD display, relay, and buzzer. During operation, the voltage sensor measures battery voltage, the current sensor measures charging/discharging current, and the temperature sensor monitors battery temperature.

The microcontroller continuously converts analog sensor signals into digital values using its internal ADC. The measured parameters are processed to estimate battery State of Charge (SOC) and determine battery condition.

If all measured values remain within predefined safe operating limits, the LCD continuously displays battery information, and the relay maintains normal battery operation.

Whenever battery voltage exceeds the maximum limit, falls below the minimum limit, current exceeds the allowable value, or temperature rises beyond the safety threshold, the Arduino immediately disconnects the battery using the relay and activates the buzzer to alert the user.

This continuous monitoring cycle ensures safe battery operation while preventing damage caused by abnormal operating conditions.

4.9 Hardware Components

The hardware required for implementing the proposed system includes:

- Arduino Uno (ATmega328P)
- Voltage Sensor Module
- Current Sensor (ACS712)
- Temperature Sensor (LM35)
- 16×2 LCD Display (I2C)
- Relay Module (5V)
- Buzzer
- Potentiometer (Simulation)
- Battery / DC Power Supply
- Breadboard
- Jumper Wires
- USB Cable

Each component performs a specific function in monitoring and protecting the battery system.

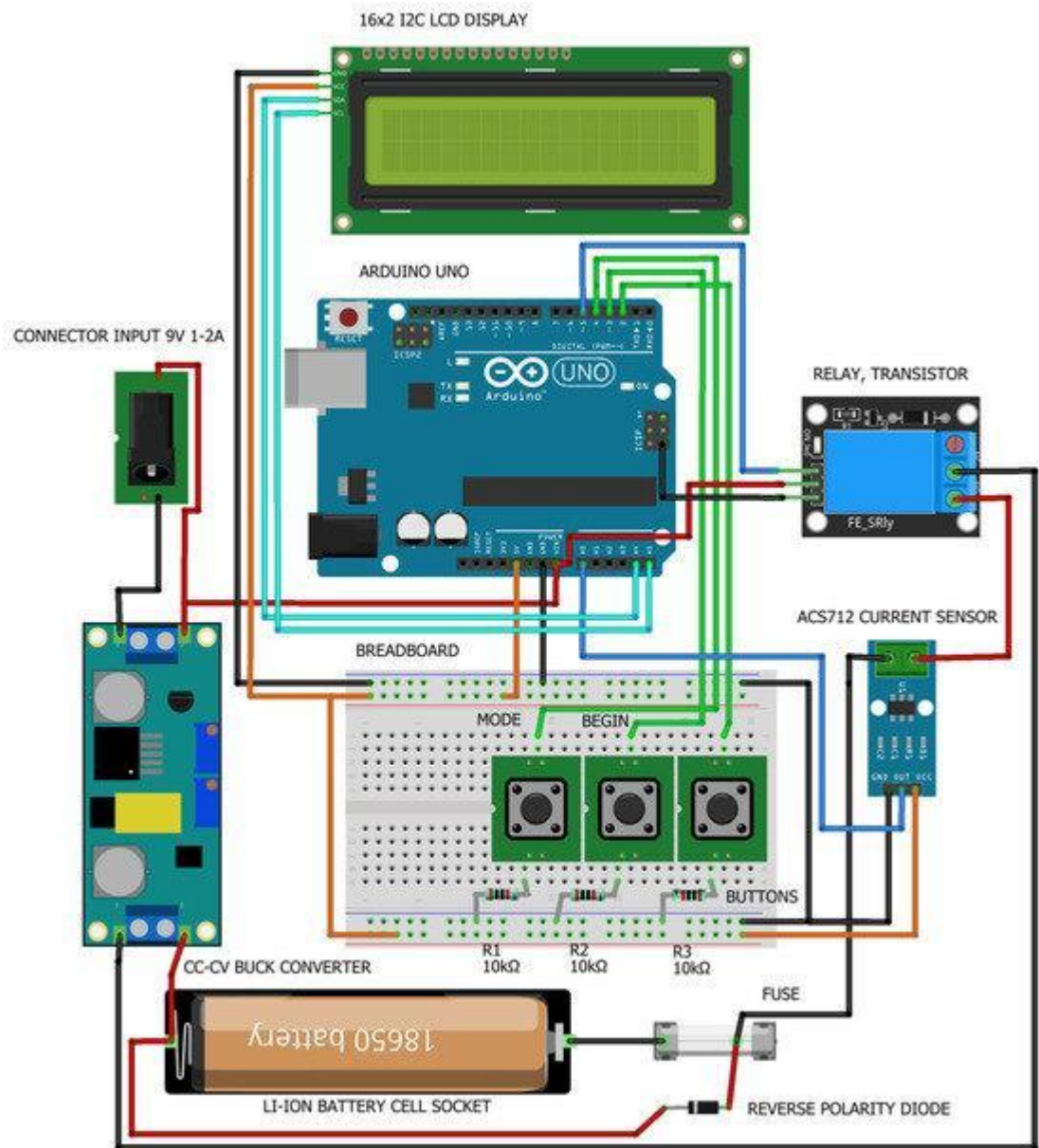


Figure 4.3: Hardware Setup of the Proposed System

4.10 Software Requirements

The software tools used for implementing the project include:

- Arduino IDE
- Embedded C Programming
- Wokwi Simulator
- Tinker cad (for initial testing)

The Arduino IDE is used for writing, compiling, debugging, and uploading the embedded program into the Arduino Uno.

4.11 Flowchart

The operational flow of the system is as follows:

1. Start System
2. Initialize Arduino and Sensors
3. Read Voltage, Current, and Temperature
4. Calculate SOC
5. Display Battery Parameters
6. Compare with Safety Limits
7. If Normal → Continue Monitoring
8. If Fault Detected → Activate Relay and Buzzer
9. Repeat Monitoring Process

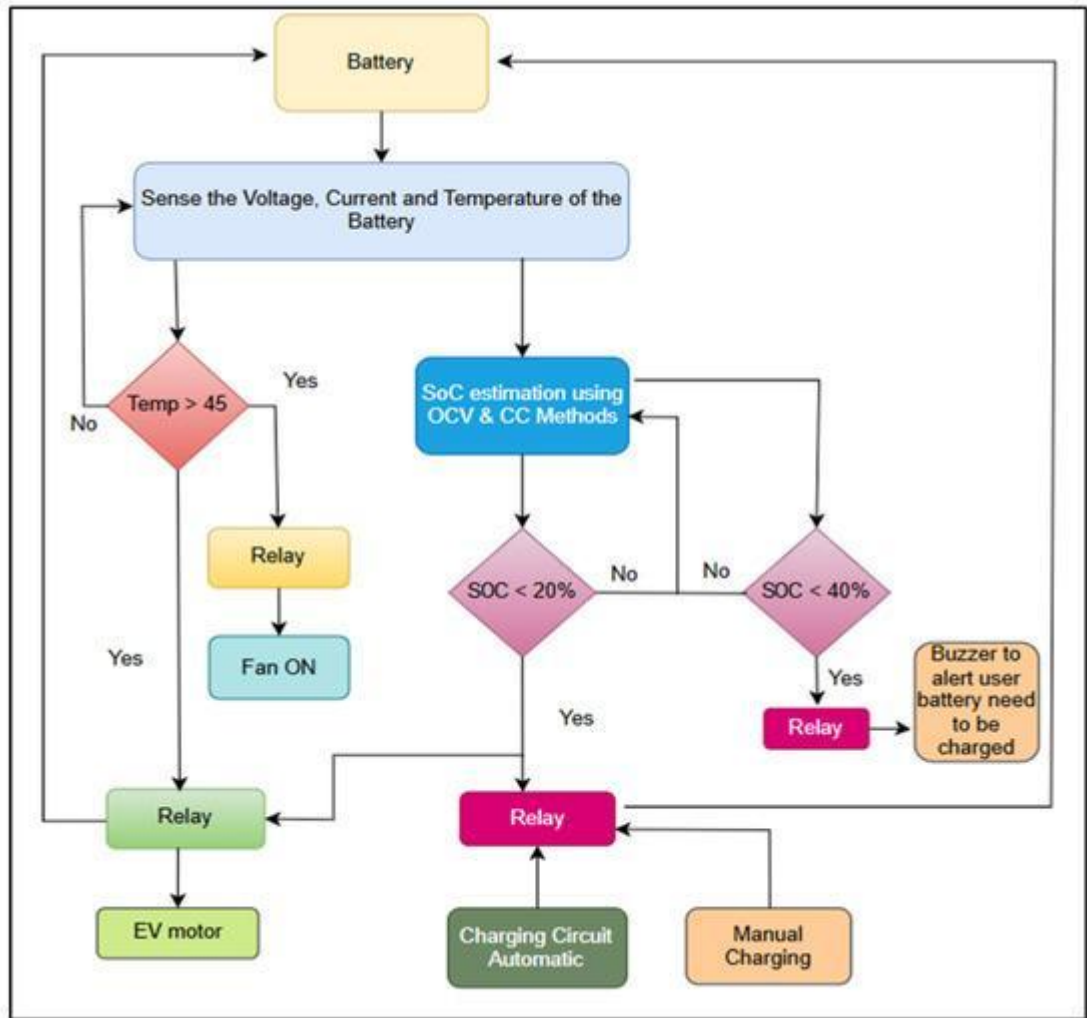


Figure 4.2: Flowchart of the Proposed System

4.12 Advantages

The proposed Smart EV Battery Monitoring and Protection System offers several advantages:

- Continuous real-time battery monitoring.
- Automatic battery protection against faults.
- Improved battery lifespan.
- Reduced maintenance cost.
- Enhanced battery safety.

- Simple hardware implementation.
- Low development cost.
- Easy to program and maintain.
- Expandable for IoT applications.
- Suitable for educational and prototype projects.

4.13 Applications

The developed system can be applied in:

- Electric Vehicles
- Electric Scooters
- Solar Battery Storage Systems
- UPS Battery Monitoring
- Electric Bicycle Battery Packs
- Portable Power Stations
- Battery Charging Stations
- Research Laboratories
- Engineering Colleges
- Embedded Systems Training

4.14 Summary

This chapter introduced the proposed Smart EV Battery Monitoring and Protection system, including its problem statement, objectives, need, scope, hardware and software requirements, block diagram, working principle, advantages, and applications. The proposed system integrates real-time monitoring with automatic protection to

CHAPTER 5

PROJECT IMPLEMENTATION

CHAPTER 5

PROJECT IMPLEMENTATION

5.1 Introduction

The implementation phase is the most important stage of the project, where the proposed Smart EV Battery Monitoring and Protection system is developed by integrating both hardware and software components. The primary objective of this implementation is to continuously monitor the battery's voltage, current, temperature, and State of Charge (SOC) while ensuring the battery operates within safe limits. The system is designed using an Arduino Uno microcontroller, which acts as the central processing unit responsible for collecting sensor data, processing battery parameters, making protection decisions, and controlling the output devices.

The hardware implementation involves connecting sensors, display modules, relays, and alarms to the Arduino Uno. The software implementation consists of developing an Embedded C program using the Arduino IDE that reads sensor values, performs calculations, displays battery information, and activates protection mechanisms whenever abnormal conditions occur. The complete system has been tested using the Wokwi simulation platform before practical implementation to verify its functionality.

The developed prototype demonstrates a simple, economical, and efficient Battery Management System suitable for educational purposes, prototype development, and small-scale electric vehicle applications.

5.2 Hardware Design

The hardware design consists of the following major components:

- Arduino Uno (ATmega328P)
- Voltage Sensor Module
- ACS712 Current Sensor
- LM35 Temperature Sensor
- I2C 16×2 LCD Display
- Relay Module

- Buzzer
- Potentiometers (for simulation in Wokwi)
- Battery / DC Supply
- Breadboard and Jumper Wires

The Arduino Uno receives analog input signals from the voltage, current, and temperature sensors. These values are converted into digital signals using the built-in Analog-to-Digital Converter (ADC). The processed information is displayed on the LCD display while the relay and buzzer operate according to the protection logic implemented in the software.

5.3 Circuit Diagram.

The voltage sensor measures the battery voltage and is connected to analog pin A0 of the Arduino. The ACS712 current sensor is connected to analog pin A1 to measure charging and discharging current. The LM35 temperature sensor is connected to analog pin A2 for continuous temperature monitoring.

The I2C LCD display communicates with the Arduino through SDA and SCL pins. The relay module is connected to digital pin D8, while the buzzer is connected to digital pin D9. The Arduino is powered using a regulated 5V power supply.

The circuit ensures reliable communication between sensors and the controller while providing automatic battery isolation during fault conditions.

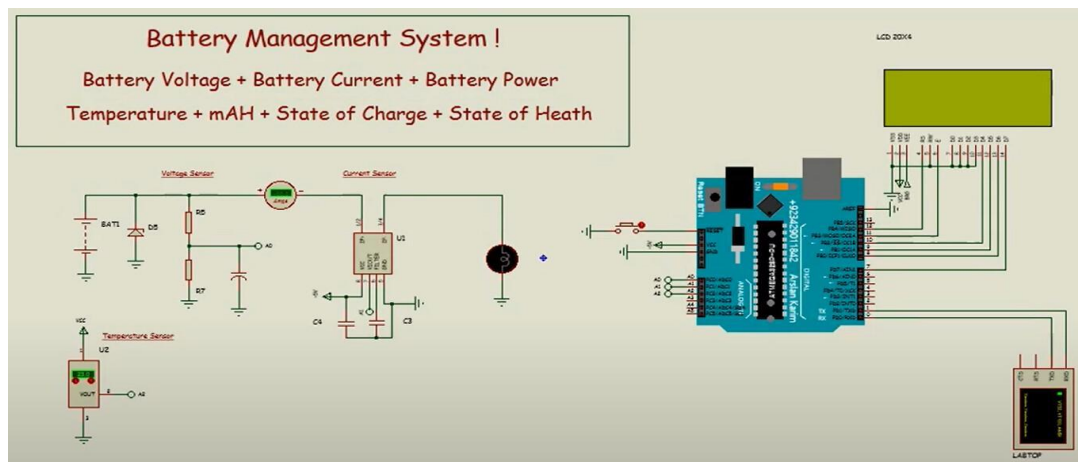


Figure 5.1: Circuit Diagram of the Proposed System

5.4 Hardware Connections

The hardware connections are summarized below:

Component	Arduino Pin
Voltage Sensor	A0
ACS712 Current Sensor	A1
LM35 Temperature Sensor	A2
LCD SDA	A4
LCD SCL	A5
Relay Module	D8
Buzzer	D9
5V Supply	5V
Ground	GND

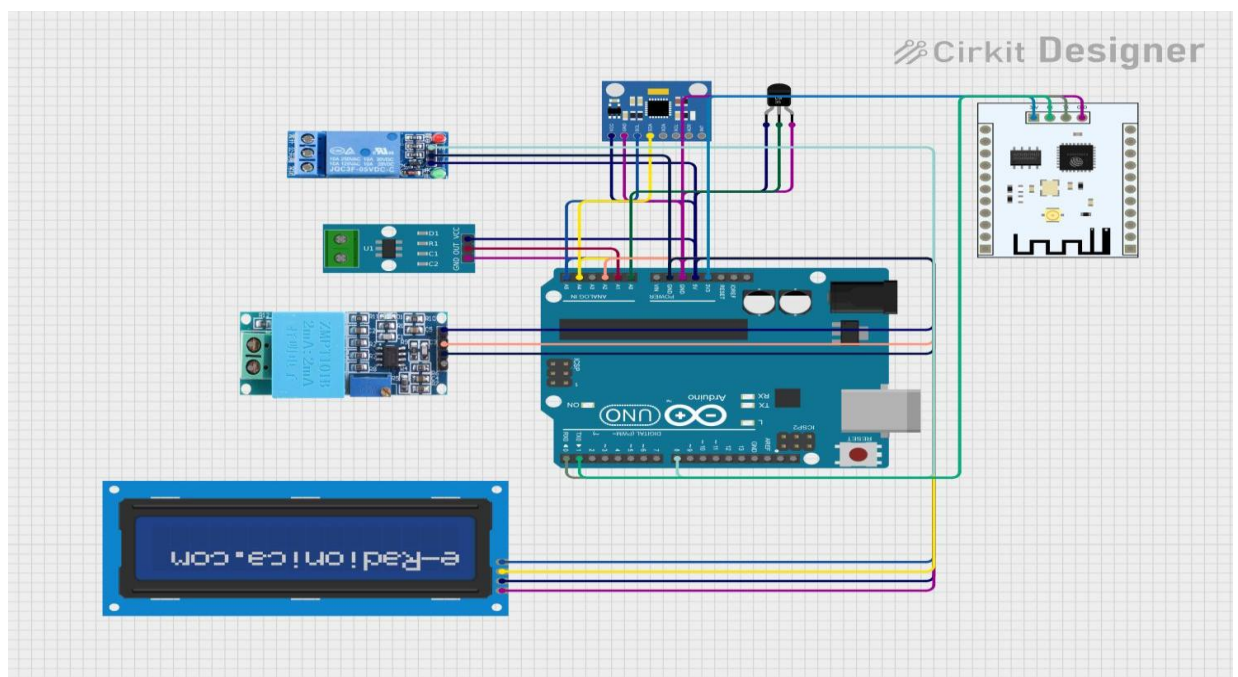


Figure 5.2: Hardware Connections of the Implemented Prototype.

5.5 Software Design

The software is developed using the Arduino IDE in Embedded C. The program is structured into two main functions

Setup() Function

The setup function initializes the LCD display, serial communication, sensor pins, relay, and buzzer. It executes only once when the Arduino is powered on or reset.

Loop() Function

The loop function continuously performs the following operations:

- Reads voltage sensor value.
- Reads current sensor value.
- Reads temperature sensor value.
- Calculates State of Charge (SOC).
- Displays battery parameters.
- Compares sensor values with safety thresholds.
- Controls relay and buzzer.
- Repeats the monitoring process continuously.

This cyclic execution enables real-time battery monitoring.

5.6 Program Flow

The software algorithm follows the sequence below:

1. Initialize Arduino.
2. Initialize LCD.
3. Configure relay and buzzer.
4. Read sensor values.
5. Convert analog values into engineering units.
6. Calculate battery SOC.
7. Display battery parameters.
8. Compare with threshold values.
9. If battery is healthy → Continue monitoring.
10. If fault occurs → Disconnect relay.
11. Activate buzzer.
12. Continue monitoring.

This logical flow ensures uninterrupted battery supervision.

Project Name: Management System with Charge Monitoring and Fire Protection.

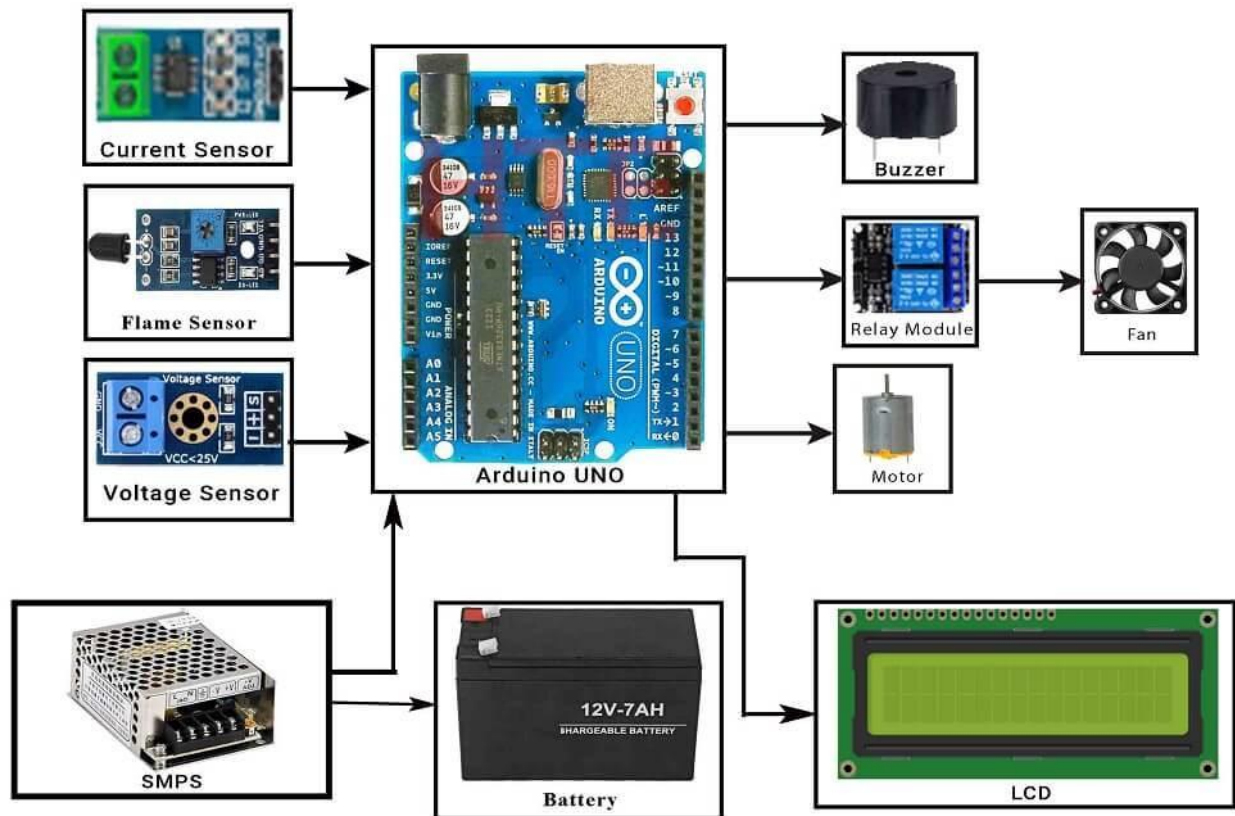


Figure 5.3: Software Flowchart for Battery Monitoring and Protection

5.7 Arduino Program Explanation

The Arduino program is responsible for coordinating all system operations.

Initially, the necessary libraries for the LCD display are included. Global variables are declared for voltage, current, temperature, and SOC calculations.

Inside the setup function, the LCD display is initialized, and all digital output pins are configured. During continuous execution, the Arduino reads analog values from the sensors, converts them into meaningful engineering values, and calculates the battery State of Charge.

Threshold values are defined for voltage, current, and temperature. Whenever any parameter exceeds its safe operating limit, the relay is switched OFF to disconnect the battery, and the buzzer generates an audible warning. If all values remain within safe limits, the relay remains ON, allowing normal battery operation.

5.8 Battery Monitoring Algorithm

The monitoring algorithm performs continuous acquisition and processing of battery parameters.

The voltage sensor measures the instantaneous battery voltage. The current sensor measures charging and discharging current flowing through the battery. The LM35 temperature sensor continuously measures battery temperature.

The measured values are processed to determine the battery's operating condition. The Arduino estimates the battery State of Charge (SOC) using voltage-based calculations suitable for prototype applications.

The LCD continuously updates battery information, enabling users to observe battery health in real time.

5.9 Protection Algorithm

The protection algorithm ensures battery safety by comparing measured parameters with predefined safety limits.

The following protection conditions are implemented:

- Overvoltage Protection – Relay disconnects the battery if voltage exceeds the maximum limit.
- Undervoltage Protection – Relay disconnects the battery if voltage falls below the minimum limit.
- Overcurrent Protection – Relay disconnects the battery if charging/discharging current exceeds the safe value.
- Overtemperature Protection – Relay disconnects the battery if temperature exceeds the predefined threshold.

Whenever a fault is detected:

- Relay is turned OFF.
- Buzzer is activated.
- Warning message is displayed on LCD.

The system automatically resumes normal operation after parameters return within safe limits.

5.10 LCD Display Implementation

The LCD provides a user-friendly interface for monitoring battery status.

The following parameters are displayed:

- Battery Voltage (V)
- Battery Current (A)
- Battery Temperature (°C)
- State of Charge (SOC %)
- Battery Status (Normal / Charging / Fault)

The display updates continuously, allowing users to monitor battery conditions without additional equipment.

5.11 Relay Control

The relay acts as the battery protection switch.

During normal operating conditions, the relay remains energized, connecting the battery to the load or charger.

Whenever abnormal conditions occur, the Arduino de-energizes the relay, disconnecting the battery instantly to prevent damage.

The relay automatically reconnects the battery when safe operating conditions are restored.

5.12 Testing Procedure

The implemented system was tested using both simulation and practical methods.

Initially, the complete circuit was verified in the Wokwi simulator. Potentiometers were used to simulate voltage, current, and temperature variations.

Various fault conditions were intentionally introduced to evaluate system performance.

The following scenarios were tested:

- Normal battery operation.
- Overvoltage condition.
- Low battery voltage.
- High charging current.
- Excessive battery temperature.
- Relay switching.
- Buzzer operation.
- LCD parameter display.

All tests confirmed the correct functioning of the monitoring and protection algorithms.

5.13 Challenges Faced

During project implementation, several challenges were encountered.

Sensor calibration required repeated adjustments to obtain accurate voltage, current, and temperature readings. Initial relay switching caused unstable outputs due to incorrect threshold settings. LCD display timing required optimization to avoid screen flickering. SOC estimation was simplified because advanced algorithms require precise battery models and extensive computational resources.

Despite these challenges, the system was successfully implemented and demonstrated stable operation under different testing conditions.

5.14 Summary

This chapter described the complete implementation of the Smart EV Battery Monitoring and Protection system. It covered the hardware design, circuit connections, software architecture, monitoring algorithm, protection algorithm, LCD interface, relay control, and testing procedure. The integration of sensors, Arduino Uno, LCD display, relay module, and buzzer resulted in a reliable prototype capable of continuously monitoring battery parameters and protecting the battery from unsafe operating conditions. The next chapter presents the experimental results, performance analysis, and evaluation of the developed system.

CHAPTER 6

RESULTS

CHAPTER 6

RESULTS

6.1 Introduction

The results obtained from the implementation of the Smart EV Battery Monitoring and Protection system demonstrate its ability to monitor critical battery parameters and provide automatic protection under abnormal operating conditions. The project was successfully implemented using an Arduino Uno microcontroller, voltage sensor, ACS712 current sensor, LM35 temperature sensor, relay module, buzzer, and a 16×2 LCD display. The system was initially tested in the Wokwi simulation environment before validating the functionality through hardware implementation.

The experimental results confirm that the proposed system accurately measures battery voltage, current, temperature, and State of Charge (SOC). The LCD continuously displays these parameters in real time, allowing users to monitor battery health effectively. Whenever any parameter exceeds its predefined safety limit, the relay disconnects the battery and the buzzer generates an audible warning, ensuring battery safety and preventing damage.

The obtained results verify that the proposed system performs continuous monitoring, provides reliable protection, and responds quickly to abnormal battery conditions. The implementation demonstrates the effectiveness of combining embedded systems with sensor technology to develop an economical Battery Management System suitable for educational and prototype electric vehicle applications.

6.2 Experimental Setup

The experimental setup consisted of an Arduino Uno connected to a voltage sensor, ACS712 current sensor, LM35 temperature sensor, I2C LCD display, relay module, and buzzer. Potentiometers were used in the Wokwi simulation to vary voltage, current, and temperature values for testing different battery conditions.

The Arduino continuously acquired sensor data, processed battery parameters, displayed information on the LCD, and controlled the relay and buzzer according to the implemented

protection algorithm. Different operating conditions were created by adjusting the simulated sensor values to verify the performance of the monitoring and protection system.

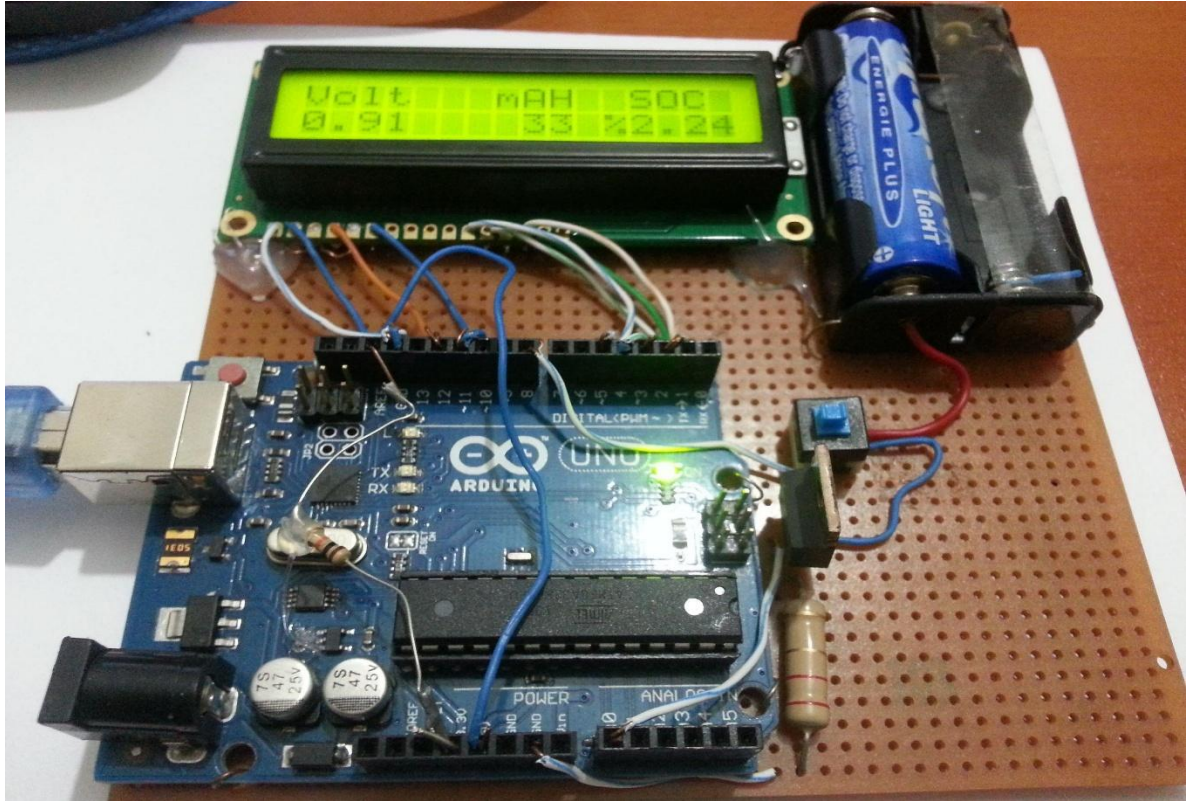


Figure 6.1: LCD Display Showing Battery Parameters

6.3 Battery Voltage Monitoring Results

The voltage monitoring circuit successfully measured battery voltage throughout charging and discharging operations. Under normal conditions, the measured voltage remained within the predefined operating range and was continuously displayed on the LCD.

When the battery voltage exceeded the maximum threshold, the relay immediately disconnected the battery from the charging circuit. Similarly, when the voltage dropped below the minimum safe limit, the relay isolated the battery to prevent deep discharge. The response time was sufficiently fast to avoid unsafe battery operation.

The voltage monitoring results confirmed the accuracy and reliability of the implemented sensing and protection mechanism.

6.4 Current Measurement Results

The ACS712 current sensor continuously measured charging and discharging current flowing through the battery. During normal operation, the measured current remained within acceptable limits, allowing uninterrupted battery operation.

When excessive current conditions were simulated, the Arduino immediately detected the abnormal current level and activated the relay protection circuit. The buzzer simultaneously alerted the user regarding the fault condition.

The current monitoring system effectively prevented excessive current flow, thereby improving battery safety and protecting connected electronic components.

6.5 Temperature Monitoring Results

Battery temperature was continuously monitored using the LM35 temperature sensor. Under normal operating conditions, the measured temperature remained within the safe operating range, and the LCD displayed the real-time battery temperature.

Whenever the battery temperature exceeded the predefined threshold value, the protection algorithm immediately disconnected the battery using the relay while simultaneously activating the warning buzzer.

The temperature monitoring system effectively prevented battery overheating, thereby reducing the possibility of thermal damage and improving overall battery reliability.

6.6 State of Charge (SOC) Results

The State of Charge (SOC) represents the percentage of energy remaining in the battery. The Arduino estimated SOC using voltage-based calculations appropriate for prototype implementation.

During charging, the SOC value increased gradually until the battery reached full charge. During discharge, the SOC value decreased according to battery usage. The calculated SOC

was continuously displayed on the LCD, allowing users to estimate the remaining battery capacity.

The implemented SOC estimation method provided satisfactory results for demonstration purposes and successfully reflected battery charging and discharging conditions.

6.7 Relay Protection Results

The relay module functioned as the primary protection device within the proposed Battery Management System.

The relay remained energized during normal battery operation. Whenever any of the following fault conditions occurred:

- Overvoltage
- Undervoltage
- Overcurrent
- Overtemperature

the Arduino immediately de-energized the relay, disconnecting the battery from the load or charging source. Once battery parameters returned to normal operating limits, the relay restored normal system operation.

The relay protection mechanism demonstrated reliable and consistent performance throughout all testing scenarios.

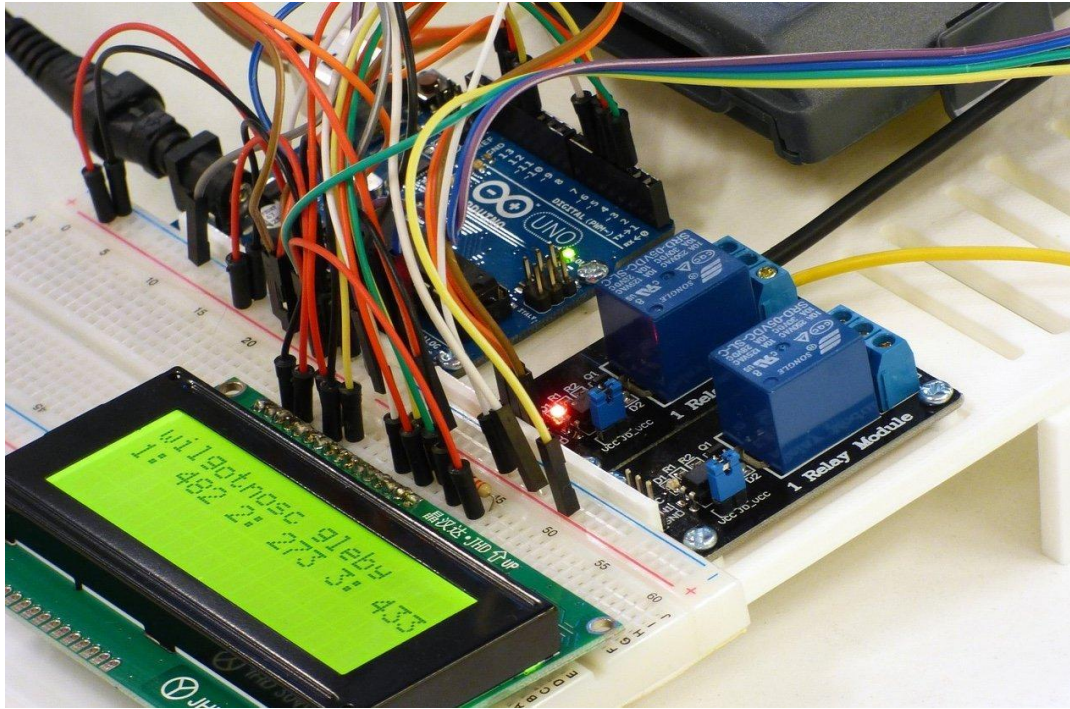


Figure 6.2: Relay Protection During Fault Condition

6.8 LCD Display Results

The 16×2 LCD display continuously presented battery information to the user in an easy-to-read format. The displayed parameters included:

- Battery Voltage
- Battery Current
- Battery Temperature
- State of Charge (SOC)
- Battery Status

The display updated automatically whenever sensor values changed. During fault conditions, warning messages such as "OVER VOLTAGE," "HIGH TEMP," or "FAULT DETECTED" were displayed to inform the user of the battery status.

The LCD interface significantly improved system usability by providing real-time battery information.

6.9 Performance Analysis

The implemented Battery Monitoring and Protection System achieved reliable performance under all tested operating conditions. Sensor measurements were stable, and the Arduino successfully processed battery parameters without noticeable delay.

The relay responded immediately whenever abnormal battery conditions were detected, while the buzzer effectively alerted the user. The LCD continuously updated battery information without interruption.

The overall system demonstrated good monitoring accuracy, rapid fault detection, and dependable protection performance. The implementation also confirmed that Arduino-based Battery Management Systems provide an economical solution for educational and prototype applications.

6.10 Comparison with Existing Systems

Parameter	Existing Systems	Proposed System
Battery Voltage Monitoring	Yes	Yes
Current Monitoring	Limited	Yes
Temperature Monitoring	Yes	Yes
State of Charge Display	Limited	Yes
LCD Display	Optional	Yes
Relay Protection	Limited	Yes
Buzzer Alert	No	Yes
Low Cost	No	Yes

Parameter	Existing Systems	Proposed System
Arduino-Based	No	Yes
Easy Implementation	Moderate	Yes

The comparison indicates that the proposed system provides comprehensive monitoring and protection while maintaining low implementation cost and simplicity.

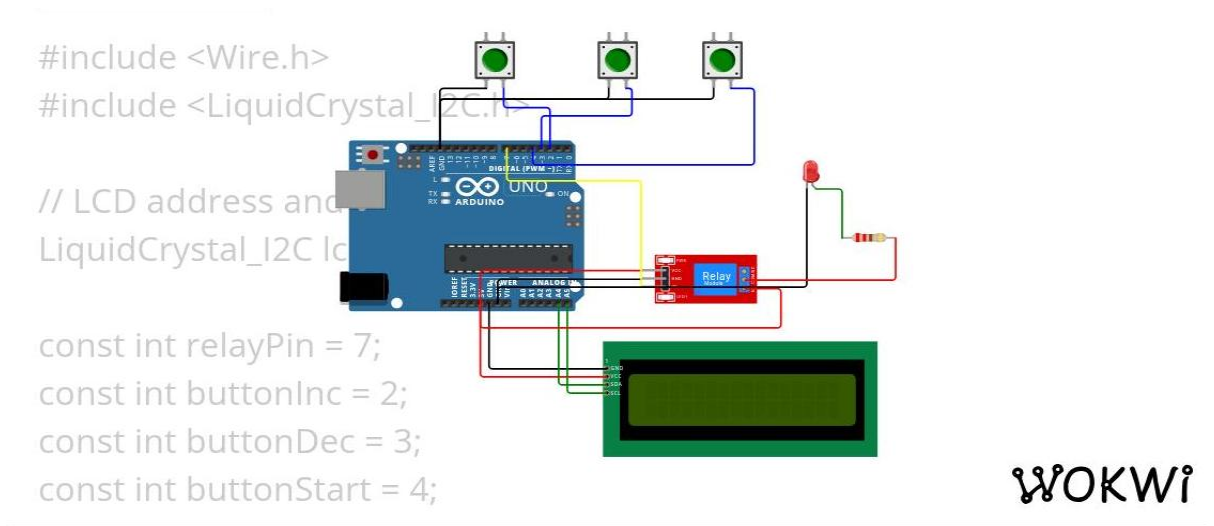


Figure 6.3: Wokwi Simulation Results of the Smart EV Battery Monitoring System

6.11 Advantages Achieved

The developed Smart EV Battery Monitoring and Protection system successfully achieved the following advantages:

- Continuous real-time battery monitoring.
- Automatic protection against battery faults.
- Improved battery safety.
- Increased battery lifespan.
- Reliable relay-based protection.
- Immediate fault indication through buzzer.
- User-friendly LCD display.

- Low hardware cost.
- Easy maintenance.
- Expandable for IoT integration.

These advantages demonstrate the practical usefulness of the developed prototype.

6.12 Summary

The experimental results confirmed the successful implementation of the Smart EV Battery Monitoring and Protection system. The Arduino-based prototype accurately monitored battery voltage, current, temperature, and State of Charge while providing automatic relay-based protection during abnormal operating conditions. The LCD display provided continuous real-time battery information, and the buzzer effectively alerted users whenever faults occurred. Overall, the developed system demonstrated reliable performance, low implementation cost, ease of operation, and suitability for educational, research, and prototype electric vehicle applications. The final chapter presents the overall conclusion of the project along with possible future enhancements.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

CHAPTER 7

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7.1 Conclusion

The Smart EV Battery Monitoring and Protection system was successfully designed, implemented, and tested using an Arduino Uno microcontroller, voltage sensor, ACS712 current sensor, LM35 temperature sensor, relay module, buzzer, and 16×2 LCD display. The developed system effectively monitored the battery's voltage, current, temperature, and State of Charge (SOC) in real time while providing automatic protection against abnormal operating conditions. The integration of embedded systems with sensor technology enabled continuous battery supervision and improved operational safety.

During experimental testing, the proposed system accurately measured battery parameters and displayed them on the LCD display. Whenever battery voltage exceeded the maximum limit, dropped below the minimum limit, current exceeded the safe operating range, or temperature increased beyond the predefined threshold, the Arduino immediately disconnected the battery through the relay and activated the buzzer to alert the user. These protection mechanisms successfully prevented unsafe battery conditions that could otherwise reduce battery life or cause permanent damage.

The implementation demonstrated that a low-cost microcontroller-based Battery Management System can effectively perform real-time battery monitoring and protection without requiring expensive commercial hardware. The developed prototype is simple to construct, easy to program, and highly suitable for educational institutions, research laboratories, prototype electric vehicle systems, renewable energy storage applications, and embedded systems training.

Overall, the project successfully achieved all the objectives established at the beginning of the work. It provides an economical, reliable, and scalable solution for intelligent battery monitoring while demonstrating the practical application of embedded systems in electric vehicle technology. The proposed system contributes to safer battery operation, improved energy utilization, enhanced battery lifespan, and increased system reliability.

7.2 Project Achievements

The major achievements of the project are summarized below:

- Successfully designed and developed a Smart EV Battery Monitoring and Protection system.
- Implemented continuous monitoring of battery voltage, current, temperature, and State of Charge (SOC).
- Developed an automatic relay-based battery protection mechanism.
- Integrated a buzzer for real-time fault indication.
- Displayed battery parameters continuously using a 16×2 LCD display.
- Successfully simulated the complete system using the Wokwi simulation platform.
- Verified the functionality of the monitoring and protection algorithms under different operating conditions.
- Achieved a low-cost, simple, and reliable Battery Management System suitable for prototype implementation.
- Demonstrated the application of embedded systems in electric vehicle battery management.

7.3 Limitations

Although the developed prototype successfully performs battery monitoring and protection, certain limitations remain due to the scope of the project.

The implemented State of Charge (SOC) estimation is based on voltage measurements and therefore provides approximate values rather than highly accurate battery capacity estimation. Advanced commercial Battery Management Systems generally employ algorithms such as Coulomb Counting, Extended Kalman Filters, or Artificial Intelligence techniques to improve estimation accuracy.

The proposed system monitors a single battery pack and does not perform individual cell monitoring or cell balancing, which are essential for large lithium-ion battery packs used in commercial electric vehicles. Additionally, remote monitoring through IoT platforms and cloud-based data storage has not been implemented in the current version.

Despite these limitations, the developed system effectively demonstrates the fundamental principles of Battery Management Systems and provides an excellent platform for future enhancements.

7.4 Future Scope

The developed Battery Monitoring and Protection System offers several opportunities for future improvements to meet the requirements of commercial electric vehicle applications.

Future versions of the system may incorporate **IoT-based remote monitoring**, allowing users to observe battery parameters using smartphones or web applications from any location. Cloud integration can enable long-term storage of battery data, performance analysis, and predictive maintenance.

Advanced **State of Health (SOH)** estimation techniques may be incorporated to evaluate battery aging and predict remaining battery life. Artificial Intelligence and Machine Learning algorithms can also be integrated for intelligent fault diagnosis, battery life prediction, and adaptive charging control.

Future research may include **battery cell balancing**, which ensures uniform charging and discharging among individual battery cells, thereby improving battery efficiency and lifespan. Wireless Battery Management Systems (WBMS) can further reduce wiring complexity while improving reliability.

The system may also be integrated with **GPS modules** to provide vehicle location tracking, **GSM/Wi-Fi communication modules** for wireless alerts, and **mobile applications** that display real-time battery status. Renewable energy integration, fast-charging management, and cloud-based fleet monitoring are additional areas that can significantly enhance the capabilities of the proposed system.

These improvements would transform the prototype into a comprehensive intelligent Battery Management System suitable for commercial electric vehicles, energy storage systems, and smart transportation networks.

7.5 Summary

This chapter presented the overall conclusion of the **Smart EV Battery Monitoring and Protection** project and summarized its major achievements. The developed system successfully demonstrated real-time battery monitoring, automatic protection, and reliable embedded control using the Arduino Uno microcontroller. The project confirmed that continuous monitoring of battery parameters significantly improves battery safety, reliability, and operational lifespan while reducing the risk of battery failures. Furthermore, several future enhancement opportunities such as IoT integration, Artificial Intelligence, cloud computing, battery health estimation, and wireless communication were discussed, providing a roadmap for developing a more advanced and intelligent Battery Management System in the future.

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Use the **IEEE referencing style**, which is commonly required for B.Tech engineering project reports.

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